PHYSICS

1) An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}(V_0 > 0)$ enters an electric field $\vec{E} = -E_0\hat{i}$ (E = constant > 0) at t = 0. If λ_0 is its de-Broglie wavelength initially, then its de- Broglie wavelength at time t is :

(1) $\frac{\lambda_0}{\left(1+\frac{eE_0}{mV_0}t\right)}$

(2) $\lambda_0\left(1+\frac{eE_0}{mV_0}t\right)$

(3) $\lambda_0 t$

(4) λ_0

2) When an electron in hydrogen atom is excited from its 4th to 5th stationary orbit, the change in angular momentum of electron is :-

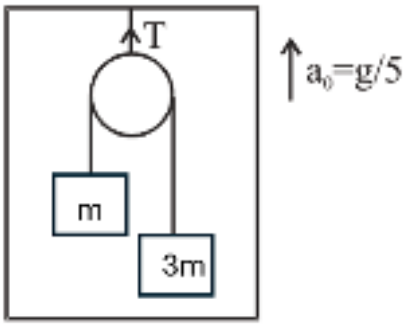
(1) 4.6×10^{-34} J.s

(2) 3.32×10^{-34} J.s

(3) 1.05×10^{-34} J.s

(4) 2.08×10^{-34} J.s

3) If the system is going upwards with acceleration $a_0 = g/5$, find the tension 'T' shown in figure.



(1) $\frac{18\,mg}{5}$

(2) $\frac{10\,mg}{5}$

(3) $\frac{mg}{5}$

(4) $\frac{6\,mg}{5}$

4) Two particles 'A' and 'B' start SHM at t = 0. Their positions as a function of time are given by $X_A = A \sin \omega t$; $X_B = A \sin (\omega t + \pi/3)$

	List-I		List-II
(P)	Minimum time when x is same	(a)	$\frac{5\pi}{6\omega}$
(Q)	Minimum time when velocity is same	(b)	$\frac{\pi}{3\omega}$
(R)	Minimum time after which $v_A < 0$ and $v_B < 0$	(c)	$\frac{\pi}{\omega}$
(S)	Minimum time after which $x_A < 0$ and $x_B < 0$	(d)	$\frac{\pi}{2\omega}$

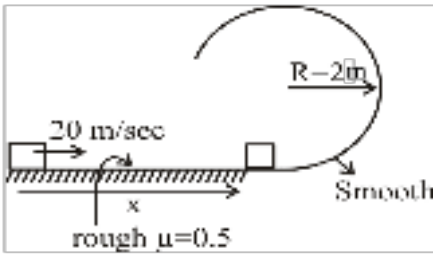
(1) P → a; Q → b; R → d; S → c

(2) P → b; Q → a; R → d; S → c

(3) P → b; Q → a; R → c; S → d

(4) P → c; Q → a; R → d; S → b

5) A block of mass m is given a velocity 20 m/sec on horizontal rough surface of length x m. After rough surface their is a vertical circular path of radius 2m which is frictionless. The block enters on the circular path with minimum velocity so that it can complete the path. What is the length of rough surface x if coefficient of friction is μ = 0.5.



(1) 30 m

(2) 20 m

(3) 10 m

(4) 16 m

6)  In given hollow cylindrical conductor current density is $J = \frac{J_0}{r^2}$, where J_0 is constant and r is the distance from axis of cylinder. If radius of inner surface is a and radius of outer surface is 2a then find current passed through the conductor parallel to axis.

(1) $\pi J_0 \log 2$

(2) $2\pi J_0 \log 2$

(3) $2\pi J_0$

(4) $2\pi J_0 \log 3$

7) The moment of inertia of a body about a given axis is 2.4 kg-m^2. Time taken to produce a rotational kinetic energy of 750 J, an angular acceleration of 5 rad/s^2 must be applied about that axis for :

(1) 6 sec

(2) 5 sec

(3) 4 sec

(4) 3 sec

8) Two waves represented by the following equations are travelling in the same medium $y_1 = 5\sin 2\pi (75t - 0.25x)$, $y_2 = 10 \sin 2\pi (150t - 0.50x)$. The intensity ratio I_1/I_2 of the two waves is:

(1) 1 : 2

(2) 1 : 4

(3) 1 : 8

(4) 1 : 16

9) In a resonance tube experiment, the tube resonates with tuning fork when it is filled with water upto a height of 21.0 cm from bottom. When water level is gradually raised, same tuning fork gives next resonance at a height of 58.5 cm. If speed of sound at room temperature is 330 ms^{-1}, the frequency of tuning fork is given by :

(1) 360 Hz

(2) 440 Hz

(3) 620 Hz

(4) 880 Hz

10) A uniform rod of mass m, length L is rotating in horizontal plane about axis passing through center and perpendicular to length with angular velocity ω. If a particle of mass m/2 is gently attached to one of its end, then new angular velocity will be-

(1) 2ω / 5

(2) ω / 3

(3) 3ω / 5

(4) ω / 2

11) Choose the correct statements about a p-n junction diode :
(a) New holes and conduction electrons are produced continuously through out the material.
(b) The potential barrier across the depletion region is due to ions.
(c) Barrier potential of a PN junction depends upon, type of biasing, temperature and doping density.
(d) A millivoltmeter connected across an open PN junction terminal can only measure built in barrier potential.

(1) a & b

(2) a, c & d

(3) a, b & c

(4) c only

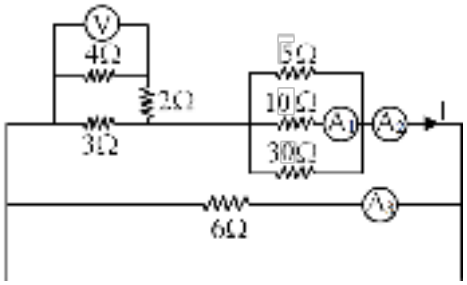
12) A vessel open at top contains 50 litres of water. A small opening is made at the bottom of vessel. It is observed that 3 litres of water comes out in time t_1, the next 3 litres in a further time t_2 and the next 3 litres in further time t_3, then :-

(1) $t_1 = t_2 = t_3$

(2) $t_1 < t_2 < t_3$

(3) $t_1 > t_2 > t_3$

(4) $t_1 = t_2 < t_3$

13) 

For the circuit shown above, if the reading of voltmeter V is 8V, the readings of ammeter A_1, A_2 and A_3 in the above circuit will be respectively.

(1) 0.9A, 12A, 8A

(2) 0.9 A, 6A, 3A

(3) 1.8A,6A, 5A

(4) 2.7A,5A, 6A

14) In a new system, 1 unit of time is equal to 10s, 1 unit of mass is 5 kg and 1 unit of length is 20m. In this new system of units, 1 unit of energy is equal to :-

(1) 20J

(2) $\frac{1}{20}$J

(3) 4J

(4) 16J

15) The fraunhofer diffraction pattern of a single slit is formed at the focal plane of a lens of focal length 1m. The width of the slit is 0.3 mm. If the third minimum is formed at a distance of 5 mm from the central maximum then calculate the wavelength of light.

(1) 3000Å

(2) 4000Å

(3) 5000Å

(4) 6000Å

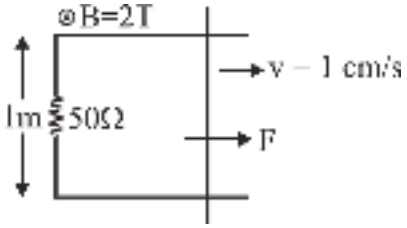
16) **Assertion :** When battery remains connected, electric potential energy will increase if dielectric material is inserted between plates of capacitor.
Reason : When battery remains connected, charge on plates of capacitor remains same.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

17) 

A wire is sliding on two parallel conducting rails placed at a separation of 1m as shown in figure. Magnetic field 2T exists inwards perpendicular to rails. What force is necessary to keep the wire moving with a constant velocity of 1 cm/sec?

(1) 0.08×10^{-3} N

(2) 8×10^{-6} N

(3) 4×10^{-6} N

(4) 8×10^{-4} N

18) A man can swim in still water at a speed of 5 km/hr. He wants to cross a river 6 km wide, flowing at the rate of 4 km/hr. If he heads in a direction making an angle of 127° with stream direction, then he will reach a point on the other bank :

(1) upstream at a distance of 1.5 km

(2) downstream at a distance of 1.5 km

(3) directly on the other side of the bank

(4) never reach the other bank

19) In the photoelectric effect experiment the collector is kept at 1.5 volt with respect to emitter. A light of wavelength 400 nm incident on a metal of work function 1.8 eV, then maximum & minimum kinetic energies of emitted photoelectrons on reaching collector will be respectively :-

(1) 2.8 eV, 1.5 eV

(2) 3.8 eV, 3 eV

(3) 5.2 eV, 2 eV

(4) 4.3 eV, 1.1 eV

20)

The height of the point vertically above the earth's surface, at which acceleration due to gravity becomes 1% of its value at the surface is (Radius of the earth = R)

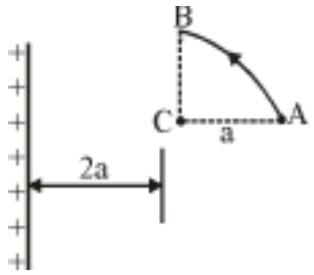
(1) 8 R

(2) 9 R

(3) 10 R

(4) 20 R

21) The arc AB with the centre C and the infinitely long wire having linear charge density λ are lying in the same plane. The minimum amount of work to be done to move a point charge q_0 from point A



to B through a circular path AB of radius a is equal to :-

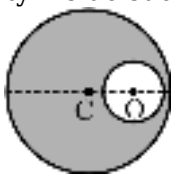
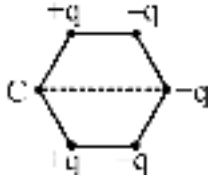
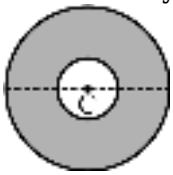
(1) $\frac{q_0\lambda}{2\pi\varepsilon_0}\log\left(\frac{2}{3}\right)$

(2) $\frac{q_0\lambda}{2\pi\varepsilon_0}\log\left(\frac{3}{2}\right)$

(3) $\frac{q_0\lambda}{4\pi\varepsilon_0}\log\left(\frac{2}{3}\right)$

(4) $q_0\,\lambda/\sqrt{2}\,\pi\varepsilon_0$

22) List-II has conditions on electric field and list-I has certain charge configurations. Match the suitable conodition(s) with each charge configurations.

List-I		List-II	
(P)	Two point charges +q and –q are placed along x-axis and C is a point nearer to +q 	(1)	Electric field at the point C is towards right (positive x-direction)
(Q)	A uniformly positively charged sphere with a sperical cavity inside such that centre C and O lie on x-axis 	(2)	Electric field at the point C is zero.
(R)	A regular hexagon with 5 charges as shown. Dotted line shows x-axis. 	(3)	Electric field at the point C is towards left (negative x direction)
(S)	A uniformly negatively charged sphere with a sperical cavity inside such that centre of cavity coincides with centre of sphere C 	(4)	Electric field at the point C has two component one along negative x and other along positive y.

(1) P → 1; Q → 3; R → 3; S → 2

(2) P → 1; Q → 1; R → 3; S → 2

(3) P → 1; Q → 2; R → 4; S → 2

(4) P → 3; Q → 2; R → 4; S → 2

23) A sphere of radius 10 cm is charged to a potential of 3000 V. The outward pull per unit area of its surface is :-

(1) 4×10^{-3} N/m^2

(2) 2×10^{-3} N/m^2

(3) 1×10^{-3} N/m^2

(4) Zero

24) ***Assertion (A)*** **:-** In series L-C-R circuit, voltage function will lead the current function for frequency greater than the resonance frequency.
Reason (R) **:-** At resonance frequency, phase difference between current function and voltage function is zero.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

25) Match list-I with list-II

	List-I		**List-II**
(a)	Rectifier	(i)	Used for stepping up or stepping down the A.C. voltage
(b)	Stabilizer	(ii)	Used to convert A.C. voltage into DC voltage
(c)	Transformer	(iii)	Used to remove any ripple in the rectified output voltage
(d)	Filter	(iv)	Used for constant output voltage

(1) a-(ii), b-(iv), c-(i), d-(iii)

(2) a-(iii), b-(iv), c-(i), d-(ii)

(3) a-(ii), b-(i), c-(iv), d-(iii)

(4) a-(ii), b-(i), c-(iii), d-(iv)

26) The root mean square speed of hydrogen molecules of an ideal hydrogen gas kept in a gas chamber is 3180 m/s. The pressure of the hydrogen gas is :-
(Density of hydrogen gas = 8.99×10^{-2} kg/m^3, 1 atmosphere $= 1.01 \times 10^5$ N/m^2)

(1) 1.0 atmosphere

(2) 1.5 atmosphere

(3) 2.0 atmosphere

(4) 3.0 atmosphere

27) Consider a planet of mass M revolving round the sun in an elliptical orbit with period T and angular momentum L. Then the area of the elliptical orbit will be

(1) $\frac{L}{2M}$

(2) $\frac{L}{2M}T$

(3) $\frac{L}{2M}T^2$

(4) $\frac{L}{M}T$

28)

The position vector of a particle is given as $\vec{r} = \left(t^2 - 4t + 6\right)\hat{i} + \left(t^2\right)\hat{j}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to :-

(1) 1 sec.

(2) 2 sec.

(3) 1.5 sec.

(4) 5 sec.

29) A convex lens is placed on optical bench between fixed distance of object and screen. When lens is slowly moved on optical bench, away from object, then at a certain position it forms a sharp real image of size 16 cm. When lens is further displaced by 30 cm in same direction along principle axis, it again forms a sharp real image of size 4 cm on screen. The focal length of the lens is :

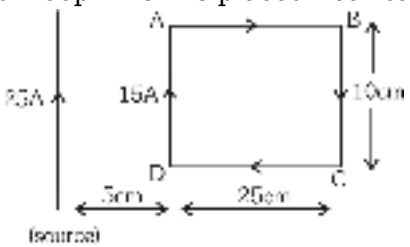
(1) 15 cm

(2) 20 cm

(3) 24 cm

(4) 48 cm

30) A rectangular loop ABCD is placed near to infinite length current carrying wire. Magnetic force on the loop is :-



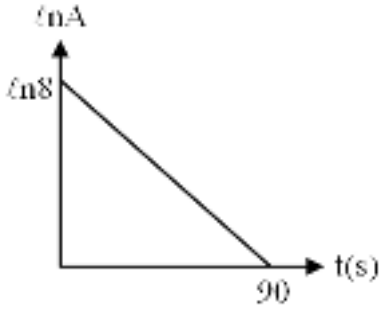
(1) 1.25×10^{-4} N, Attraction

(2) 1.25×10^{-4} N, Repulsion

(3) 12.5×10^{-4} N, Repulsion

(4) 12.5×10^{-4} N, Attraction

31) In study of damped oscillation of simple pendulum, if A denotes amplitude and t time, the graph between nA versus t is as shown in figure. The time in which amplitude reduces to 4 cm from initial



amplitude of 8 cm is

(1) 15 s
(2) 30 s
(3) 45 s
(4) 60 s

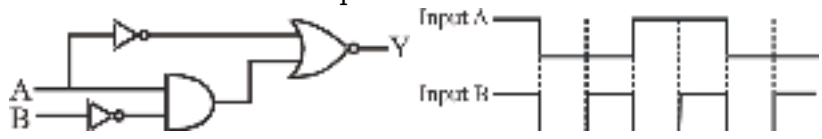
32) Binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.47 MeV. The energy required to remove a neutron from C^{13} is :-

(1) 5.49 MeV
(2) 4.95 MeV
(3) 9.45 MeV
(4) 5.94 MeV

33) Magnetic field at the centre of a circular loop of area A is B. Then magnetic moment of the loop will be :-

(1) $\frac{BA^2}{\mu_0 \pi}$
(2) $\frac{BA}{\mu_0}\sqrt{A}$
(3) $\frac{BA\sqrt{A}}{\mu_0 \pi}$
(4) $\frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$

34) The logic circuit shown has the input waveforms 'A' and 'B' as shown. Pick out the correct output waveform :-



(1)



(2)



(3) 

(4) 

35) A cylindrical region of radius 1.2m is containg a uniform electric field that is increasing uniformly with time. At t = 0 the field is zero and at t = 5.0 s the field is 200 V/m. The total displacement current through a cross section of the region is :-

(1) 4.5×10^{-16}A

(2) 2.0×10^{-15} A

(3) 3.5×10^{-10} A

(4) 1.6×10^{-9} A

36) One mole of an ideal diatomic gas at temperature T_0 expands slowly according to law P = KV (K is a constant). If final temperature is $3T_0$ heat supplied to gas is -

(1) $2RT_0$

(2) $6RT_0$

(3) $4RT_0$

(4) $3RT_0$

37) Two open organ pipes give 4 beats/sec when sounded together in their fundamental nodes. If the length of the pipe are 100 cm and 102.5 cm respectively, then the velocity of sound is:

(1) 496 m/s

(2) 328 m/s

(3) 240 m/s

(4) 160 m/s

38) A metallic cube whose each side is 10 cm is subjected to a shearing force of 2100 N. The top face is displaced through 0.25 cm with respect to bottom. The shear modulus will be :-

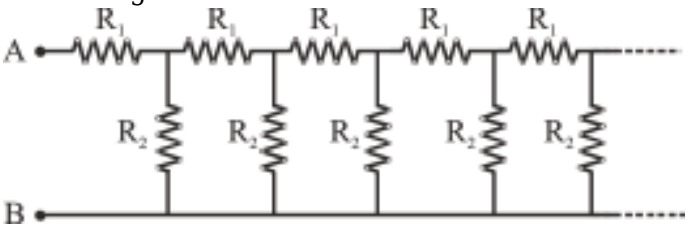
(1) 8.4×10^{6} N/m^2

(2) 525×10^{-4} N/m^2

(3) 52.5×10^{4} N/m^2

(4) 5.25×10^{7} N/m^2

39) An infinite sequence of resistance is shown in the figure. The resultant resistance between A and B will be, when R_1 = 1 ohm and R_2 = 2 ohm :-



(1) Infinity

(2) 1Ω

(3) 2Ω

(4) 1.5 Ω

40) Two particles of mass 2 kg and 1 kg are moving along the same straight line with speeds 2 m/s and 5 m/s respectively. The speed of COM of system when particles are moving in opposite directions is V_1 and when particles are moving in same direction is V_2. The ratio V_1/V_2 is :

(1) 3:4

(2) 4:3

(3) 9:1

(4) 1:9

41) In a vernier calliper, 1 cm on main scale is divided into 10 equal divisions and 10 vernier scale divisions are equal to 9 divisions of main scale. When both jaws of vernier calipers touch each other, the zero of the vernier scale is present at the left of zero of main scale and 4th vernier scale division exactly coincides with a main scale division. While measuring the diameter of a sphere. it was found that zero of the vernier scale lies between 35th and 36th division of main scale and 6th vernier division exactly coincides with a main scale division. The correct diameter of spherical body is :

(1) 3.50 cm

(2) 3.52 cm

(3) 3.62 cm

(4) 3.70 cm

42) A general body have surface area 20 m^2 and emissivity 0.8. If the emissive power of an ideal black body is 200 $\frac{watt}{m^2}$ at the same temperature, then amount of emitted energy in one minute by the general body is:-

(1) 45 kJ

(2) 192 kJ

(3) 300 kJ

(4) None of these

43) A 2 kg block slides on a horizontal floor with a speed of 4m/s. It strikes an uncompressed spring, and compresses it till the block is motionless. The kinetic friction is 15 N and spring constant is 10,000 N/m. Then spring compresses by-

(1) 5.5 cm

(2) 2.5 cm

(3) 11.0 cm

(4) 8.5 cm

44) A thin lens of refractive index 1.5 has a focal length of 15 cm in air. When the lens is placed in water of refractive index 4/3, its focal length will become :-

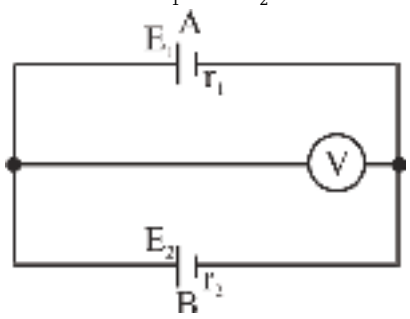
(1) 30 cm

(2) 45 cm

(3) 60 cm

(4) 75 cm

45) Two cells A and B of emf 1.3 V and 1.5 V respectively are arranged as shown in figure. Find relation between r_1 and r_2 for which the voltmeter reads 1.45 V. The voltmeter is assumed to be ideal.



(1) $r_1 = 2r_2$

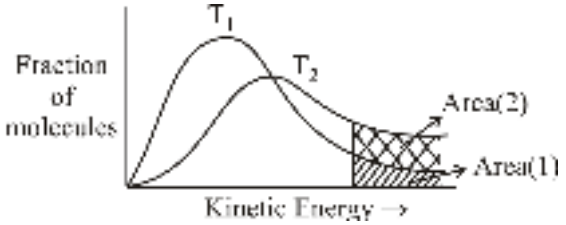
(2) $r_1 = 3r_2$

(3) $r_2 = 2r_1$

(4) $r_2 = 3r_1$

CHEMISTRY

1)



(a) $T_1 < T_2$ (T = Temperature)

(b) On increasing temperature, fraction of molecules containing most probable kinetic energy increases.

(c) On increasing temperature, value of most probable kinetic energy increases.

(d) On changing temperature area under curve remains constant since total probability must be one at all time.

The correct statements are

(1) a, c, d

(2) a, b, c, d

(3) a, c

(4) only d

2) Match the column-I & column-II :-

	Column I (Character of reaction)		**Column II (Order)**

(A)	Reaction with identical rate and rate constant	(p)	Zero
(B)	Reaction rate doubles on increasing concentration four times	(q)	First
(C)	Half-life period is inversely proportional to initial concentration	(r)	Second
(D)	Half-life period is independent of initial concentration	(s)	Half

(1) (A) → (p); (B) → (s); (C) → (r); (D) → (q)

(2) (A) → (p); (B) → (r); (C) → (s); (D) → (q)

(3) (A) → (p); (B) → (q); (C) → (r); (D) → (s)

(4) (A) → (r); (B) → (s); (C) → (p); (D) → (q)

3) Which one of the following pairs of solution can we expect to be isotonic at the same temperature ?

(1) 0.1 M urea and 0.1 M NaCl

(2) 0.1 M urea and 0.2 M $MgCl_2$

(3) 0.1 M NaCl and 0.1 M Na_2SO_4

(4) 0.1 M $Ca(NO_3)_2$ and 0.1 M Na_2SO_4

4) Read the following statements carefully and choose the correct statements.
(i) The vapour pressure of a liquid decreases with increase of temperature.
(ii) The liquid boils at the temperature at which its vapour pressure is equal to the atmospheric pressure.
(iii) Vapour pressure of the solvent decreases in the presence of non-volatile solute.
(iv) Vapour pressure of the pure solvent and solution is a function of temperature.

(1) (i), (ii) and (iv) are correct

(2) (i), (iii) and (iv) are correct

(3) (ii), (iii) and (iv) are correct

(4) (i), (ii) and (iii) are correct

5) $A^{-2} + B \rightarrow A + B^{-2}$
$B^{-2} + C \rightarrow$ No reaction
$C^{-2} + A \rightarrow C + A^{-2}$
The correct order of standard oxidation potential is–

(1) $E^0_{C^{-2}/C} > E^0_{A^{-2}/A} > E^0_{B^{-2}/B}$

(2) $E^0_{B^{-2}/B} > E^0_{C^{-2}/C} > E^0_{A^{-2}/A}$

(3) $E^0_{C^{-2}/C} > E^0_{B^{-2}/B} > E^0_{A^{-2}/A}$

(4) $E^0_{A^{-2}/A} > E^0_{B^{-2}/B} > E^0_{C^{-2}/C}$

6) Select the correct statement for lead storage battery.

(1) It consists of a grid of lead packed with PbO_2 as anode

(2) A 50% solution of sulphuric acid is used as an electrolyte

(3) When battery is in use then PbO_2 is converted to $PbSO_4$ (s) at cathode

(4) On charging the battery concentration of H_2SO_4 decreases.

7) 518 gm of an aqueous solution contains 18 gm of glucose (mol. wt. = 180). What is the molality of the solution.

(1) 0.1 m

(2) 0.2 m

(3) 0.3 m

(4) 0.4 m

8) Which of the following statement is incorrect about 1.7 g of NH_3 :-

(1) It contains 0.3 mol H-atom

(2) Mass % of hydrogen is 17.65%

(3) It contains 2.408×10^{23} atoms

(4) It contains 0.3 mol N-atom

9) The uncertainty in position for an electron is $\frac{\lambda}{4\pi}$ where λ is de-Broglie wavelength. The uncertainty in velocity will be (where v is the velocity of electron) :-

(1) $\frac{v}{2}$

(2) v

(3) 3v

(4) $\frac{v}{4}$

10) In a system $P_{(s)} \rightleftharpoons 2Q_{(g)} + R_{(g)}$, at equilibrium the concentration of 'Q' is doubled then concentration of R at new equilibrium will be :-

(1) Double of its original concentration

(2) $\frac{1}{4}$ of its original concentration

(3) $\frac{1}{2}$ of its original concentration

(4) 4 times of its original concentration

11) Which of the following sets of concentrations will cause the precipitation of $ZnCl_2$ ($K_{sp} = 1.2 \times 10^{-12}\ M^3$) ?

(1) $[Zn^{2+}] = 10^{-8}$ M ; $[Cl^-] = 10^{-8}$ M

(2) $[Zn^{2+}] = 10^{-5}$ M ; $[Cl^-] = 10^{-4}$ M

(3) $[Zn^{2+}] = 10^{-6}$ M ; $[Cl^-] = 10^{-5}$ M

(4) $[Zn^{2+}] = 10^{-5}$ M ; $[Cl^-] = 10^{-3}$ M

12) $ClO_3^- + I_2 + xH_2O \rightarrow IO_3^- + Cl^- + yH^+$ The values of x + y in balanced redox reaction will be:-

(1) 5

(2) 7

(3) 9

(4) 4

13) For the reaction, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
ΔU = – 90 kcal/mole and ΔS = –200 cal K^{-1} at 500 K. What can be said about reaction ?

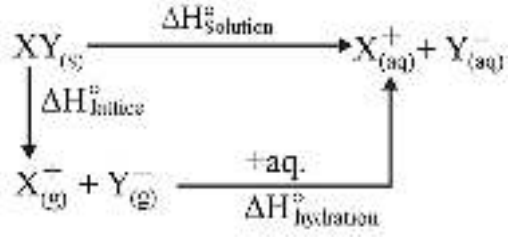
(1) Reaction will be spontaneous at all temperature and at constant pressure

(2) Reaction is at state of equilibrium

(3) Reaction is non spontaneous at all conditions

(4) Reaction is spontaneous below 460 K and at constant pressure

14) Study the figure given below :



The standard enthalpy of solution $(\Delta H^0_{solution})$ of $XY_{(s)}$ in water can be determined by -

(1) $\Delta H^0_{lattice} = \Delta H^0_{hydration} + \Delta H^0_{solution}$

(2) $\Delta H^0_{hydration} = \Delta H^0_{lattice} + \Delta H^0_{solution}$

(3) $\Delta H^0_{solution} = \Delta H^0_{lattice} + \Delta H^0_{hydration}$

(4) $\Delta H^0_{solution} = 2\Delta H^0_{hydration} - \Delta H^0_{lattice}$

15) Molar mass of Mohr's salt is 392. In Mohr's salt iron is oxidised by $KMnO_4$ in acidic medium. Equivalent weight of Mohr's salt is :

(1) 392

(2) 31.6

(3) 278

(4) 156

16) **Statement-I :** Neptunium and Plutonium like Actinium and Protactinium are found in "pitche blend" i.e., an ore of uranium.
Statement-II : A plot of $\sqrt{\nu}$ against atomic mass gave a straight line (where ν is frequency of x-

rays emitted).

(1) Both statement I and statement II are true.

(2) Both statement I and statement II are false.

(3) Statement I is true but statement II is false.

(4) Statement I is false but statement II is true.

17) Select the correct statement
(A) Ionisation enthalpy of neutral atom are always positive.
(B) The second ionisation enthalpy will be higher than the first ionisation enthalpy.
(C) Ionisation enthalpy of alkali metals are highest in each period.
(D) Ionisation enthalpy of noble gases are highest in each period.

(1) (A), (C) are correct

(2) (A), (B), (D) are correct

(3) (A), (B), (C), (D) are correct

(4) (C), (D) are correct

18) Match the following

	Column-I (Species)		**Column-II (Bond order)**
a	H_2^+	p	0
b	He_2	q	1.5
c	N_2^+	r	0.5
d	O_2^-	s	2.5

(1) a - r, b - p , c - s, d - q

(2) a - p, b - r , c - s, d - q

(3) a - r, b - p , c - q, d - s

(4) a - p, b - r , c - q, d - s

19) Which of the following molecule has single super pair :-

(1) H_2O_2

(2) C_2H_6

(3) C_2H_4

(4) N_2H_4

20) Which of the following is incorrect -

(1) Viscosity order $H_2O > CH_3OH > CH_3 - O - CH_3$

(2) Boiling point order $H_2O > HF > NH_3$

(3) Molecular weight of acetic acid is double of its molecular formula. due to dimer formation by H-bonding.

(4) In o-Salicylaldehyde inter molecular H-bonding is present.

21) **Statement I :** The larger cation and the larger size of anion, the greater the covalent character of an ionic bond.
Statement II : All the ionic bonds have some covalent character similarly all covalent bonds have some ionic character.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement II is correct but Statement I is incorrect

22) In sodium nitroprusside, the oxidation number, coordination number and EAN of iron are respectively :-

(1) +3, 6, 36

(2) +1, 6, 36

(3) +2, 5, 35

(4) +2, 6, 36

23) Which of the following is not a limitation of valence bond theory ?

(1) It does not explain the colour exhibited by coordination compounds.

(2) It does not make exact predictions regarding the tetrahedral and square planar structures of 4-coordinate complexes.

(3) It does not distinguish between weak and strong ligands.

(4) Metal atom or ion under influence of ligands can use vacant orbitals for hybridisation.

24) Correct statement is.
(A) Complex $[Pt(NH_3)_2Cl_2]$ show geometrical isomerism.
(B) Optical isomers are mirror images that can not be superimposed on one another. These are called enantiomers.
(C) Chlorophyll is a complex compound of magnesium.
(D) Ethylenediamine is a mono dentate ligand.

(1) (A), (B), (C), (D) are correct options.

(2) (A), (B), (C) are correct options.

(3) (A), (B), (D) are correct options.

(4) (A), (D) are correct options.

25) The true statement for the acids of phosphorus, H_3PO_2, H_3PO_3 and H_3PO_4 is :

(1) The order of their acidity is :
$H_3PO_4 > H_3PO_3 > H_3PO_2$

(2) H_3PO_2 acts as oxidising agent in nature.

(3) All of them are tribasic acids

(4) The geometry of phosphorus is tetrahedral in all the three above mentioned acids.

26)

Incorrect statement among the following is :

(1) Graphite has layered structure and layers are held by Vander Waal's forces.
(2) Density of diamond is more than that of graphite.
(3) C_{60} contain 12 hexagonal rings and 20 pentagonal rings.
(4) Graphite's used as a dry lubricant.

27) Choose the incorrect match

(1) Kernite = $Na_2B_4O_7.4H_2O$
(2) Borax = $Na_2B_4O_7.8H_2O$
(3) Cryolite = Na_3AlF_6
(4) Phosphinic acid = H_3PO_2

28) Which elements in 3d-series shows highest & lowest enthalpy of atomization respectively.

(1) V, Zn
(2) Cr, Zn
(3) Cr, Mn
(4) Mn, Sc

29) The common oxidation state shown by actinoids is +3 with the other oxidation states being +4, +5, +6 and +7. The number of oxidation states shown by the following elements decrease in the order :
A. Np (Z = 93)
B. Am (Z = 95)
C. Ac (Z = 89)
D. Pa (Z = 91)
Choose the **correct** answer from the options given below :

(1) A > B > C > D
(2) B > A > D > C
(3) D > B > A > C
(4) A > B > D >C

30)

Match **List-I** with **List-II** :

	List-I (Test/reagent)		**List-II (Radical identified)**
A.	Lake Test	I.	NO_3^-
B.	Nessler's Reagent	II.	Fe^{3+}
C.	Potassium thiocyanate	III.	Al^{3+}

D.	Brown Ring Test	IV.	NH_4^+

Choose the **correct** answer from the options given below :

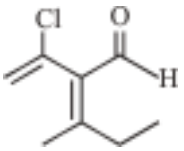
(1) A-IV, B-II, C-III, D-I

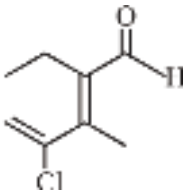
(2) A-II, B-IV, C-III, D-I

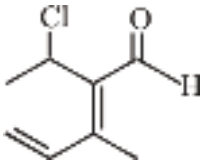
(3) A-II, B-III, C-IV, D-I

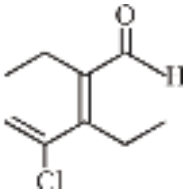
(4) A-III, B-IV, C-II, D-I

31) Structure of 4-Chloro-2-ethyl-3-methylpenta-2, 4-dienal is :

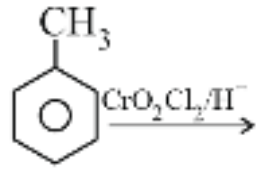
(1) 

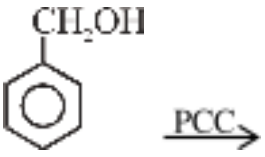
(2) 

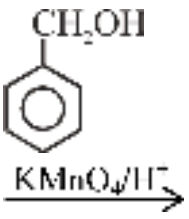
(3) 

(4) 

32) Benzaldehyde is not produced in :-

(1) 

(2) 

(3) 

(4) 

33) Which of the following statement is correct ?

(1) Aniline does not undergo friedel craft reaction due to salt formation with aluminum chloride.

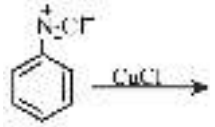
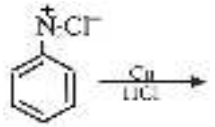
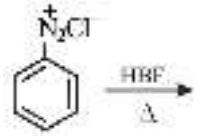
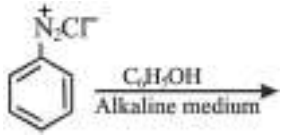
(2) Direct nitration of aniline yields tarry oxidation products in addition to the nitro derivatives.

(3) Aniline on nitration gives substantial amount of m-nitroaniline due to formation of anilinium ion.

(4) All of the above

34)

Match the column I with column II and mark the appropriate choice :-

Column I		Column II	
(A)		(i)	Coupling reaction
(B)		(ii)	Balz-Schiemann reaction
(C)		(iii)	Gatterman reaction
(D)		(iv)	Sandmeyer reaction

(1) (A)→(iv), (B)→(iii), (C)→(ii), (D)→(i)

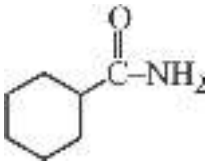
(2) (A)→(iii), (B)→(iv), (C)→(i), (D)→(ii)

(3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)

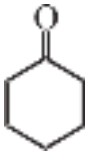
(4) (A)→(i), (B)→(ii), (C)→(iii), (D)→(iv)

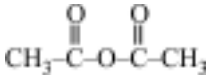
35)

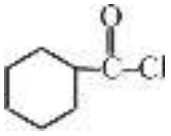
Which of the following compound will reduces by both $LiAlH_4$, as well as $NaBH_4$

(1) 

(2) 

(3) 

(4) 

(5) 

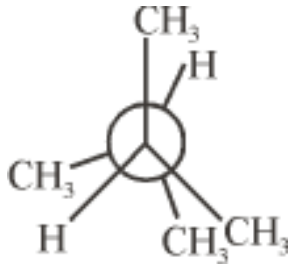
(1) 1, 3, 4

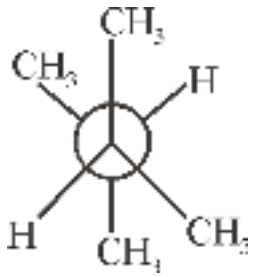
(2) 1, 5, 3

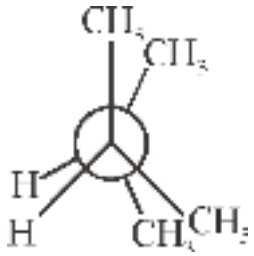
(3) 3, 5

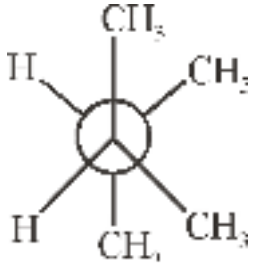
(4) 4, 5, 2

36) In which of the following has minimum torsional strain and minimum Vanderwaal strain ?

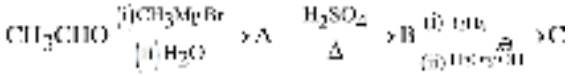
(1) 

(2) 

(3) 

(4) 

37) Compound A and C in the following reaction are :-



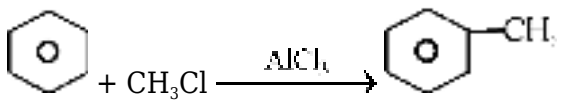
(1) Identical

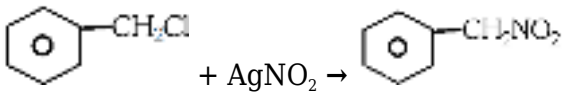
(2) Positional Isomer

(3) Functional Isomer

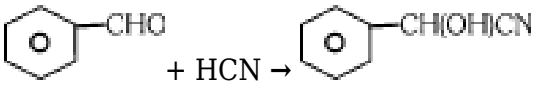
(4) Optical Isomer

38) Which one of the following is a free radical substitution :-

(1) 

(2) 

(3) 

(4) 

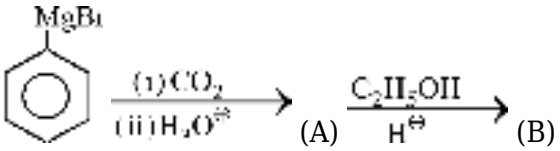
39) For denaturation of protein which statement is not correct ?

(1) Denaturation occurs by change in pH and temperature

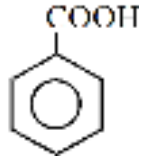
(2) Denaturation leads to loss of primary structure of protein

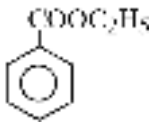
(3) Denaturation leads loss of secondary and tertiary structure of protein

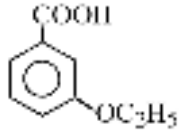
(4) Denaturation does not lead to change in sequence of amino acids in protein

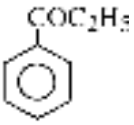
40)  (A) (B)

Identify B :-

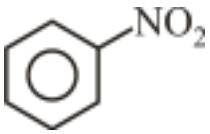
(1) 

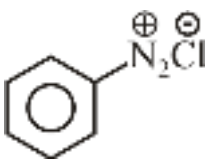
(2) 

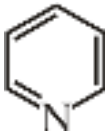
(3) 

(4) 

41) Which of the following compound will be suitable for Kjeldahl's method for nitrogen estimation ?

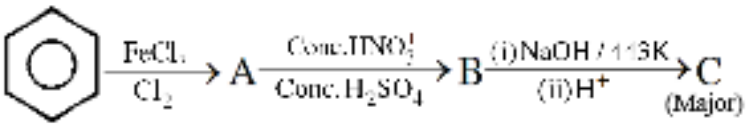
(1) 

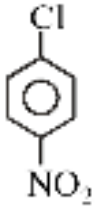
(2) 

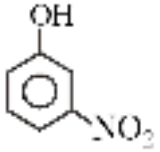
(3) 

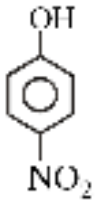
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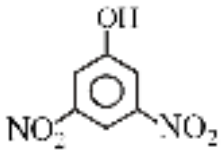
42) For the given reaction major product will be :



(1) 

(2) 

(3) 

(4) 

43) Which of the following reaction can't form higher alkane as a product.

(1) $CH_3 - Cl \xrightarrow[\text{Dry ether}]{Na}$

(2) $CH_3 - CH_2 - COONa \xrightarrow{\text{electrolysis}}$

(3) $CH_3{-}Cl \xrightarrow[2.CuCl]{1.Li}$

(4) $CH_3{-}Cl \xrightarrow[HCl]{Zn}$

44) Which of the following compounds is optically active :-

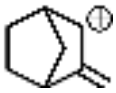
(1) n-butyl alcohol

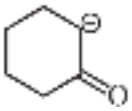
(2) Isobutyl alcohol

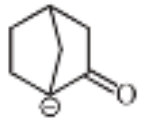
(3) Sec. butyl alcohol

(4) Ter. butyl alcohol

45) Which of the following intermediate is not stablised by resonance?

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) **Statement-I :** Injecting the microbes deliberately during immunisation induces passive immunity.
Statement-II : Directly injecting the preformed antibodies for quick response as in tetnus is type of active immunity.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

2) Given below are pairs of pathogen, diseases and type of causatic agents category. Choose the option with correctly matched.

	Disease		**Pathogen**		**Type of causatic agent**
A.	Pneumonia	(i)	Rhinovirus	(a)	Protozoan
B.	Ascariasis	(ii)	Entamoeba Histolytica	(b)	Virus
C.	Amoebiasis	(iii)	Round worm	(c)	Helminthic
D.	Common cold	(iv)	Haemophilus Influenzae	(d)	Bacteria

(1) A-iv-a, B-ii-c, C-iii-d, D-i-b

(2) A-ii-a, B-i-b, C-iv-c, D-iii-d

(3) A-iv-d, B-iii-c, C-ii-a, D-i-b

(4) A-iii-c, B-iv-a, C-i-b, D-ii-d

3) Which of the following statement is correct regarding AIDS/HIV ?

(1) It progressively decreases the number of B-Lymphocytes in the body of infected person.

(2) AIDS is caused by HIV, a member of group of uneveloped retroviruses enclosing the RNA genome.

(3) When HIV enters into cell then infected cell could survive while viruses are being replicated as in macrophage.

(4) There is always a time lag usually 2-12 week between the infection and appearance of AIDS symptoms.

4) Due to genetic or other unknown reason when body attacks on self cell this phenomenon is known as

(1) Immunological memory

(2) Contact inhibition

(3) Autoimmunity

(4) Immunisation.

5) Which of the following option is not correct.

(1) During sewage treatment in primary settling tank heterotrophic microbes consume the major part of organic matter present in sewage.

(2) Treatment of waste water is done by the heterotrophic microbes naturally present in the sewage.

(3) BOD is a measure of the organic matter present in the water.

(4) The Ministry of Environment and Forest has initiated Ganga Action Plan and Yamuna Action Plan.

6) Match the column-A with column-B and select correct option.

	Column-A		**Column-B**
(a)	*Azotobacter*	(I)	Biocontrol agents
(b)	Baculovirus	(II)	Biogas
(c)	Methanogens	(III)	Biofertilisers
(d)	Statins	(IV)	Bioactive molecule

(1) a-I, b-III, c-II, d-IV

(2) a-I, b-II, c-III, d-IV

(3) a-III, b-I, c-II, d-IV

(4) a-III, b-I, c-IV, d-II

7) Which of the following is produced by a bacteria?

(1) Roquefort cheese

(2) Swiss cheese

(3) Wine

(4) Penicillin

8) Which of the following is an example of homologous organ ?

(1) Eyes of octopus and mammal
(2) Flippers of penguin and dolphin
(3) Thorn of Bougainvillea and Tendril of Cucurbita
(4) Potato and sweet potato

9) Which of the following comes in the evolutionary history of mammals ?

(1) Synapsids, Therapsids
(2) Thecodont, Synapsids
(3) Sauropsids, Synapsids
(4) Thecodont, Saurapsids

10) Monocarpellary gynoecium with many ovules in ovary is present in :

(1) Pea
(2) Mango
(3) Maize
(4) *Hibiscus*

11) Read the following statements carefully and select the correct option from options given below :
(A) Polar nuclei are situated above the egg apparatus in embryo sac.
(B) In process of triple fusion, one male gamete fuses with one diploid female gamete.
(C) In beet and black pepper, perisperm is present in seeds.
(D) Apomixis is a type of sexual reproduction which mimics the asexual reproduction.
Options :

(1) A, C and D are correct
(2) A, B and D are correct
(3) A and C are correct
(4) Only C is correct

12) Bundle sheath is made up of parenchymatous cells in :

(1) Monocot stem
(2) Monocot leaves
(3) Dicot leaves
(4) 2 and 3 both

13)

Which of the following statement is **not** true regarding the internal structure of flowering plants?

(1) Vascular bundles are scattered in maize root
(2) Casparian strips are present in endodermis of maize root
(3) Large number of vascular bundles are arranged in ring in sunflower stem

(4) Phloem parenchyma is usually absent in maize stem

14) Match the column I with column II and select the correct option from options given below:

	Placentation		**Example**
A	Basal	P	Sunflower
B	Marginal	Q	Pea
C	Parietal	R	*Argemone*
D	Free central	S	*Dianthus*

Options :

(1) A : P; B : Q ; C : S; D : R

(2) A : P; B : Q ; C : R; D : S

(3) A : P; B : S ; C : Q; D : R

(4) A : P; B : S ; C : R; D : Q

15) One pair of simple leaves arises from same node in :

(1) Sunflower and mustard

(2) *Calotropis* and *Alstonia*

(3) *Calotropis* and guava

(4) Mustard and neem

16) How many of the following structures are present in embryo of monocot plants but not in embryo of dicot plants?
Cotyledon, embryonal axis, coleoptile, radicle, plumule, endosperm, coleorhiza, seed coat, hypocotyl, epicotyl

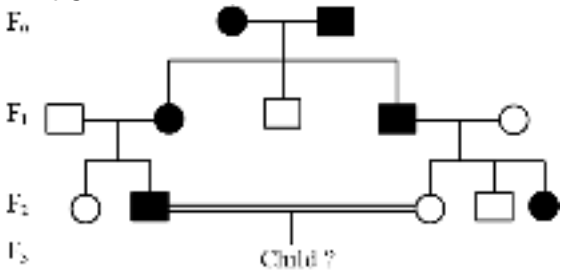
(1) Three

(2) Six

(3) Two

(4) Four

17) With the help of given pedigree find out the probability for the birth of a child having no disease in F_3 generation?



(1) $\frac{1}{2}$

(2) $\frac{2}{2}$

(3) $\frac{3}{4}$

(4) $\frac{1}{4}$

18) Match the column I with column II

	Column-I		**Column-II**
(a)	Human	(i)	XO type
(b)	Honey bee	(ii)	ZW type
(c)	Grass hopper	(iii)	XY type
(d)	Fowl	(iv)	Haploid-Diploid type

(1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

19) Which of the following statements is not correct.

(1) A true- breeding line is one that, having undergone continuous self pollination, shows the unstable trait inheritance and expression for several generations.

(2) A homozygous parent produces all gametes that are similar while a heterozygous one produces two kinds of gametes.

(3) Multiple alleles can be found only when population studies are made.

(4) Sickle-cell anaemia is an autosome linked recessive trait.

20) Given below are two statements :
Statement I : Morgan and his group found that genes of eye colour and yellow body colour in *Drosophila* were very tightly linked and showed only 1.3 per cent recombination in F_1 generation.
Statement II : Morgan carried out - several dihybrid crosses in *Drosophila* to study genes that were sex- linked.
In the light of above statement choose the correct answer from the options given below:

(1) **Statement-I** is correct but **Statement-II** is incorrect.

(2) Both **Statement-I** and **Statement-II** are incorrect.

(3) **Statement-I** is incorrect but **Statement-II** is correct.

(4) Both **Statement-I** and **Statement-II** are correct.

21) Match column-I with column-II.

	Column-I		**Column-II**
A	Secreted by Posterior pituitary	i.	Prolactin
B	Secreted by Parathyroid gland	ii.	ACTH

C	Stimulates Adrenal gland	iii.	Thyrocalcitonin
D	Secreted by Anterior pituitary	iv.	PTH
E	Secreted by Thyroid gland	v.	ADH

(1) A-v, B-iv, C-ii, D-iii, E-i

(2) A-v, B-iv, C-i, D-ii, E-iii

(3) A-iv, B-v, C-ii, D-i, E-iii

(4) A-v, B-iv, C-ii, D-i, E-iii

22) Given below are two statements one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : Catecholamines stimulate the breakdown of glycogen resulting in an increased concentration of glucose in blood.
Reason (R) : Glucagon acts mainly on the liver cells (hepatocytes) and stimulates glycogenolysis resulting in an increased blood sugar.
In the light of above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**

(3) **(A)** is correct but **(R)** is not correct.

(4) **(A)** is not correct but **(R)** is correct.

23) I act mainly at the renal tubules and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ions. Who I am ?

(1) Cortisol

(2) Secretin

(3) Aldosterone

(4) Adrenaline

24) In the following flow diagram, identify the correct labelling (A and B).



(i) A is the afferent nerve fibres transmit impulses from tissues/organs to the ANS
(ii) B is the efferent fibres transmit regulatory impulses from the CNS to the concerned peripheral tissues/ organs.
(iii) A is the afferent nerve fibres transmit impulses from tissues/ organs to the CNS.
(iv) B is the afferent fibres transmit regulatory impulses from the CNS to the concerned peripheral tissues/organs.

(1) i and iv

(2) i, ii and iv

(3) ii and iii

(4) i, ii, iii and iv

25) Brain stem includes

(1) Cerebrum, Pons, Medulla oblongata

(2) Mid brain, Pons, Cerebellum

(3) Mid brain, Pons, Medulla oblongata

(4) Thalamus, Pons, Medulla oblongata

26) The process by which seperated bands of DNA are cut out from the agarose gel and extracted from gel piece is known as

(1) Spooling

(2) Sieving

(3) Elution

(4) Precipitation

27) Read the following statements
A. Since DNA is hydrophillic molecule, it cannot pass through cell membrane.
B. Bacteria are made competent to uptake DNA by treating with monovalent cations like calcium.
C. To enable the bacteria to uptake recombinant DNA, the cells with recombinant DNA are incubated on ice followed by placing at 42°C for long period of time and back on ice.
D. In gene gun, plant cells are bombarded with high velocity microparticles of gold or tungsten coated with DNA.
Select the correct statements.

(1) A, B,C,D

(2) C,D

(3) A,D

(4) B,D

28) Read the following statements.
Statement-I : If a recombinant DNA bearing gene for resistance to Ampicillin is transferred into *E.coli* cells, the host cells become transformed into ampicillin sensitive cells.
Statement-II : If we spread above transformed cells on agar plates, containing ampicillin, only non-transformants will grow, transformed recipient cells will die.
In the light of above statements, find the correct option.

(1) **Statement (I)** and **Statement (II)** both are correct.

(2) **Statement (I)** and **Statement (II)** both are incorrect.

(3) **Statement (I)** is correct and **Statement (II)** is incorrect.

(4) **Statement (I)** is incorrect and **Statement (II)** is correct.

29) How many of the following are the applications of transgenic animals?
To understand Normal physiology and development, Study of disease, To obtain Biological products, To test safety of Vaccine, Chemical safety testing, Molecular diagnosis.

(1) 3

(2) 4

(3) 5

(4) 2

30) Read the following statements regarding nematode resistant tobacco plant. Find the correct order of sequence?
A. Parasite could not survive in a transgenic host
B. Introduction of DNA that produced both sense and anti- sense RNA in the host cells.
C. Using *Agrobacterium* vectors, nematode specific genes were introduced into host plant.
D. Two RNAs being complementary to each other formed a double stranded RNA (ds RNA) that initiated RNAi and silenced specfic mRNA of nematode.

(1) A → B → C → D

(2) C → B → D → A

(3) A → D → C → B

(4) D → B → C → A

31) Which histone protein is not present in the core of nucleosome:-

(1) H_2B

(2) H_2A

(3) H_3

(4) H_1

32) **Assertion:** During DNA replication, on one strand the replication is continuous while on the other strand it is discontinuous.
Reason : DNA Polymerase can synthesize DNA only in the 5'→ 3' direction.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

33) **Statement-I :** r-RNA carries genetic information from DNA to ribosome.
Statement-II : One t-RNA can recognize more than one codon.

(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

34)

	Column-I		**Column-II**
A.	PCR	1.	Binding site of repressor protein
B.	Operator	2.	Catalytic activity in Ribosome

C.	r-RNA	3.	RNA splicing
D.	Sn-RNA	4.	DNA amplification

(1) A-4, B-3, C-2,D-1

(2) A-4, B-1, C-3,D-2

(3) A-4, B-1, C-2,D-3

(4) A-1, B-2, C-3,D-4

35) Which of the following RNA is not required for the synthesis of protein?

(1) m-RNA

(2) Si-RNA

(3) t-RNA

(4) r-RNA

36) Find out the correct statement:-
a. RNA mostly helps in transfer and expression of information.
b. DNA was derived from RNA.
c. In eukaryotes the freshly transcribed RNA is functional, hence can directly be translated.
d. Regulation of transcription is the primary step for regulation of gene expression.

(1) a, b and c

(2) only b and d

(3) a, b and d

(4) All a, b, c and d

37) **Statement-I :** Amniocentesis is used to test for the presence of certain genetic disorders such as down syndrome, haemophilia, sickle cell anemia etc. determine survivability of the foetus.
Statement-II : In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances.
Choose the correct answer from the given options:-

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are correct.

(4) Statement-I is incorrect but statement-II is correct.

38) In the given list how many male accessory ducts are present outside testes?
Rete testis, seminiferous tubule, epididymis, vas deferens, prostate, bulbourethral gland, seminal vesicle.

(1) 2

(2) 3

(3) 4

(4) 5

39) The process of child birth is called parturition which is induced by a complex neuroendocrine

mechanism involving which of the following?

(1) GnRH
(2) Cortisol
(3) FSH
(4) Prolactin

40) Which of the following pair is not matched correctly?

(1) Periodic abstinence ⇒ Natural method
(2) Cervical caps ⇒ Prevents S.T.I
(3) Copper releasing IUD ⇒ Multiload-375
(4) Non-steroidal pill ⇒ saheli

41) Which of the following statement is false ?

(1) Mammary gland differentiates during pregnancy and secrete milk after child birth.
(2) The mammary glands are one of the female secondary sexual characteristics.
(3) The leydig cells outside the seminiferous tubules, synthesise and secrete testicular hormones (androgens).
(4) The cyclical changes in the ovary and the uterus during menstrual cycle are induced by changes in the levels of pituitary hormones only.

42) **Assertion (A) :** Each ovary is covered by a thin epithelium which encloses the ovarian stroma.
Reason (R) : Thin epithelium of ovary is made up of germ cells.
Choose the correct answer from the given options: -

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false.
(4) (A) is true but (R) is false.

43) 

Read the following statements with respect to given diagram and select the correct option.
(A) The inner compartment of the organelle is filled with a dense heterogenous substance called the matrix.
(B) The number of this organelle per cell is variable depending on the physiological activity of the cells.
(C) The inner membrane forms a number of infoldings called the cisternae towards the matrix.
(D) The matrix of this organelle also possess circular ds DNA molecule, few RNA molecules, ribosomes and the components required for the synthesis of proteins.
(E) They are called as "Power houses of the cell" as they produces cellular energy in the form of ATP by utilising light energy.
Options :-

(1) A, B and D - Correct, C and E - Incorrect

(2) A, C and E - Correct, B and D - Incorrect

(3) A, C and E - Incorrect, B and D - Correct

(4) A, B, C and E - Correct, D - Incorrect

44) The interphase nucleus has a loose and indistinct network of nucleoprotein fibre called as -

(1) Chromosome

(2) Chromatin

(3) Chromatid

(4) Kinetochore

45) Read the following statements carefully and select the **correct** statement.

(1) The axoneme usually has nine triplets of radially arranged peripheral microtubules and a pair of centrally located microtubule.

(2) Flagella are comparatively smaller than cilia and responsible for cell movement.

(3) Flagella which work like oars, causing the movement of either the cell or the surrounding fluid.

(4) Both the cilium and flagellum emerge from centriole like structure called the basal bodies.

46) Given below are the two statements : one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A) : Meiosis is the mechanism by which conservation of specific chromosome number of each species is achieved across generation in sexually reproducing organism, even though the process, per se, paradoxically, result in reduction of chromosome number by half.
Reason (R) : Meiosis involves pairing of homologous chromosomes and recombination between non sister chromatids of homologous chromosomes.

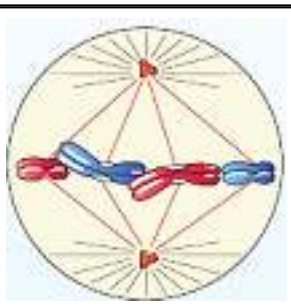
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A).

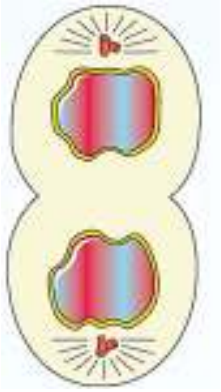
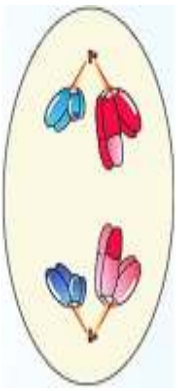
(2) (A) is true but (R) is false.

(3) (A) is false but (R) is true.

(4) Both (A) and (R) are true and (R) is correct explanation of (A).

47) Match the **column I** and **column II** and select the correct option :

	Column I		**Column II**
(A)		(I)	Homologous chromosomes separate while sister chromatids remain associated at their centromere

(B)		(II)	Spindle fibre attaches to kinetochores of chromosome
(C)		(III)	Marked by initiation of condensation of chromosomal material
(D)		(IV)	Chromosomes cluster at opposite spindle poles and their identity is lost as discrete element

(1) A-II, B-III, C-IV, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-III, B-II, C-IV, D-I

(4) A-III, B-IV, C-I, D-II

48) Given below are the two statements : one is labelled as **Assertion (A)** and other is labelled as **Reason (R)** :

Assertion (A) : Some cells in the adult animals donot appear to exhibit division (eg. heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that donot divide further exit G_1 phase to enter an inactive stage called quiscent stage (G_0) of the cell cycle.

Reason (R) : M-phase is the most drammatic period of the cell cycle, involving a major reorganization of virtually all components of cell.

In the light of above statements, choose the correct answer from the options given below -

(1) Both A and R are true but R is not the correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) Both A and R are true and R is correct explanation of A.

49) Volume of air inspired or expired per breathe during a normal respiration is.

(1) Inspiratory reserve volume

(2) Tidal volume

(3) Expiratory reserve volume

(4) Residual volume

50) The maximum volume of air a person can breathe in after a forced expiration -

(1) Expiratory capacity

(2) Vital capacity

(3) Total lung capacity

(4) Inspiratory capacity

51) A symptom of acute chest pain appears when no enough oxygen is reaching the heart muscle, is called

(1) Cardiac arrest

(2) Heart attack

(3) Angina

(4) Heart failure

52) In which of the following open circulatory system is present ?
Fishes, Arthropods, Mollusca, Amphibians Reptiles, Aves, Mammals, Annelids

(1) 3

(2) 2

(3) 4

(4) 5

53) Which of the following attributes is found in a 'Population' but NOT in an 'Individual'?

(1) Birth and Death

(2) Sex ratio and Birth rate

(3) Age

(4) Life cycle

54) Which of the following statements is **incorrect** regarding the Logistic Growth Model?

(1) It forms an 'S' shaped sigmoid curve when resources are limited.

(2) It is known as Verhulst -Pearl Logistic Growth.

(3) It occurs when resources in the habitat are unlimited.

(4) In this 'K' represents the carrying capacity.

55)

	Case		**Species involved**

(i)	Competition for resources	(1)	Barnacle - *Balanus* and *Chathamalus*
(ii)	Impact of Exotic species	(2)	Flamingoes and fishes in south American lakes
(iii)	Competitive Release	(3)	Abingdon tortoise and goats in Galapagos islands

Options :-

(1) (i) → 2, (ii) → 3, (iii) → 1

(2) (i) → 1, (ii) → 2, (iii) → 3

(3) (i) → 3, (ii) → 1, (iii) → 2

(4) (i) → 2, (ii) → 1, (iii) → 3

56) Which of the following ecosystem has the highest contribution to global net primary productivity ?

(1) Tropical deciduous forest

(2) Tropical rainforest

(3) Desert

(4) Temperate grassland

57) Which one of the following statements is **incorrect** regarding ecosystem?

(1) Net Primary Productivity (NPP) is the available biomass for herbivores and decomposers.

(2) Decomposition is slower when detritus is rich in nitrogen and water-soluble substances.

(3) In aquatic ecosystems, the grazing food chain is the major conduit for energy flow.

(4) Pyramid of energy can never be inverted.

58) **Assertion (A) :** The decomposition rate is slower when detritus contains substances like lignin and chitin.
Reason (R) : Lignin and chitin are highly resistant to microbial action and undergo very slow decomposition.
Which of the following is correct?

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

59) Choose the **incorrect** statement regarding biodiversity, its loss and its conservation.
(A) Biodiversity hotspots have high species richness and high endemism.
(B) Species richness increases with increasing area, but only up to a limit.
(C) The group Fungi has fewer species than fishes, amphibians, reptiles, and mammals combined.
(D) Habitat loss and fragmentation is the most important cause of biodiversity loss.
(E) All biodiversity hotspots together cover more than 10% of Earth's land surface.

(1) C and E only

(2) B and C only

(3) A and C only

(4) C and D only

60) **Statement A :** The world summit on sustainable development held in 2002 in Johannesburg, South Africa.
Statement B : *Ex-situ* conservation includes the establishment of wildlife sanctuaries and national parks.
Which of the following is the correct option regarding above statements ?

(1) Both Statement A and Statement B are correct.

(2) Statement A is correct, but Statement B is incorrect

(3) Statement A is incorrect, but Statement B is correct.

(4) Both Statement A and Statement B are incorrect.

61) Which of the following combinations **correctly** matches biodiversity conservation types, examples, and rationale ?

	Conservation Type	**Example**	**Rationale**
A	*In-situ*	National parks	Ecosystem protection
B	*Ex-situ*	Cryopreservation of gametes	Urgent protection of endangered species
C	Red Data Book	IUCN publication	Documentation of threatened species
D	Bioprospecting	Sacred groves	Broadly utilitarian ecological exploration

(1) A and C only

(2) A and D only

(3) A, B and C

(4) A, C and D

62) Which of the following substances are reabsorbed actively in nephron ?

(1) Glucose, Ammonia

(2) Glucose, Na^+

(3) Amino acid, Urea

(4) Na^+, Uric acid

63) Which of the following group of animals are ureotelic in nature?

(1) Human, tadpole of frog

(2) Human, frog

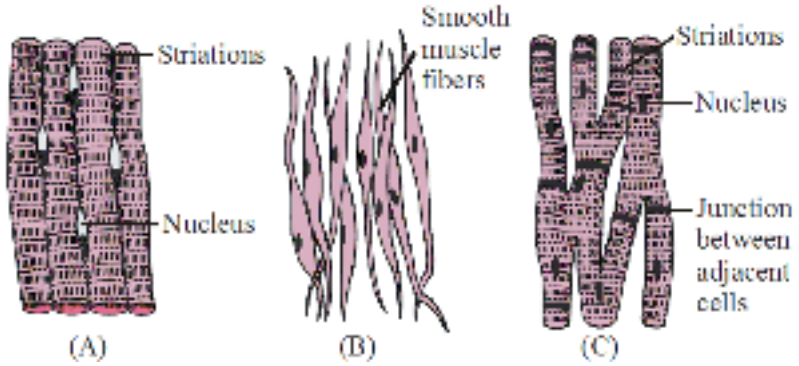
(3) Insect, frog

(4) Land snail, earthworm

64) Which is not true for scapula in the following?

(1) It is found between 2nd to 7th ribs
(2) It has an cavity which is found above the acromian process
(3) It is found dorsal part of throrax
(4) It is a flat triangular bone

65) Select the correct match :-



(1) A-Skeletal muscles → biceps, striated muscles, involuntary
(2) B-Smooth muscles → walls of stomach, voluntary
(3) C-Cardiac muscles → heart, dorsal wall of aorta, voluntary
(4) A-Skeletal muscle → biceps, striated, voluntary

66) Triploblastic, acoelomate, bilateral symmetry and most in parasitic forms. The above description is characteristic of phylum :-

(1) Protozoa
(2) Annelida
(3) Nematoda
(4) Platyhelminthes

67) Match the columns I and II, and choose the correct combination from the options given.

	Column-I		**Column-II**
(a)	Echinus	(1)	Brittle star
(b)	Cucumaria	(2)	Star fish
(c)	Antedon	(3)	Sea urchin
(d)	Ophiura	(4)	Sea cucumber
(e)	Asterias	(5)	Sea lily

(1) a – 4, b – 3, c – 5, d – 2, e – 1
(2) a – 3, b – 4, c – 5, d – 1, e – 2

(3) a – 3, b – 4, c – 5, d – 2, e – 1

(4) a – 3, b – 5, c – 4, d – 1, e – 2

68) Identify the number of organisms from the box given below which are hermaphrodites and have internal fertilisation.

Pleurobrachia, Nereis, Laccifer, Spongilla, Fasciola

(1) Six

(2) Two

(3) Four

(4) Three

69) Identify the given animal and select the incorrect option regarding this animal.



(1) Absence of blood vessels

(2) Cranium and vertebral column are cartilaginous

(3) Presence of unpaired fins

(4) Shows indirect development

70) Select the correct option in given table :

	Animal	**Character**	**Class**
1		3-Chambered heart	Reptilia
2		Operculum absent	Osteichthyes
3		Oviparous	Mammalia

4.		4- Chambered heart	Reptilia

(1) 1

(2) 2

(3) 3

(4) 4

71) Both the words in a biological name, when handwritten are, __A___, or printed in italics to indicate their___B___.

(1) A- cummulatively underline B- Latin origin

(2) A- closed in bracket B - English origin

(3) A- separately underline B- Latin origin

(4) A- cummulatively underline B- English origin

72) Match List-I and List-II.

	List -I		**List-II**
(A)	Dinoflagellates	(I)	Pigments are identical to higher plants.
(B)	Diatoms	(II)	Cell wall has stiff cellulose plates.
(C)	Euglenoids	(III)	Chief producers in the oceans
(D)	Slime moulds	(IV)	Spores possess true walls

Option :-

(1) A → III, B → II, C → IV, D → I

(2) A → II, B → III, C → IV, D → I

(3) A → III, B → IV, C → I, D → II

(4) A → II, B → III, C → I, D → IV

73) Consider the following.
(A) Chemosynthetic autotrophic bacteria oxidise various inorganic substances and use the released energy for their ATP production.
(B) In phycomycetes, zygospores are formed by meiosis and these spores are endogenously produced in sporangium.
(C) Viruses did not find a place in classification since they are not considered truly 'living'.
(D) In Lichens, the algal component is known as mycobiont and fungal component is known as phycobiont.
(E) Archaebacteria differ from other bacteria in having a different cell wall structure.
Choose the correct answer from the options given below.

(1) A, C and E are not true

(2) B and D are true

(3) A,C and E are true.

(4) A,B,C and E are true.

74) Which one of the following statement is wrong regarding gymnosperm?

(1) Male and female gametophyte do not have an independent free living existence.

(2) Multicellular female gametophyte bears two or more archegonia or female sex organs.

(3) Male gametophytic generation is highly reduced and is confined to only a limited number of cells.

(4) The megaspore mother cell divides mitotically to form four megaspore.

75) Identify A, B,C and D in the table given below.

Classes	**Major Pigments**	**Stored Food**	**Cell Wall**	**Flagellar Number and Position of Insertions**
Chlorophyceae	Chlorophyll a, b	Starch	**C**	2-8, equal, apical
Phaeophyceae	**A**	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral
Rhodophyceae	Chlorophyll a, d, phycoerythrin	**B**	Cellulose, pectin and poly sulphate esters	**D**

(1) A – Chlorophyll a, d, phycocyanin, B– Floridean starch, C–Chitin, D– 2, Short and long

(2) A – Chlorophyll a, b, fucoxanthin, B– Floridean starch,C–polysaccharide & amino acid, D–2, Longitudinally and transversely

(3) A – Chlorophyll a, d, β caroteen, B– Paramylum and fat,C–Absent, D – 2–8, heterokont flagella

(4) A – Chlorophyll a, c, fucoxanthin, B– Floridean starch, C–Cellulose, D– Absent

76) Given below are two statements :
Statement-I : In Bryophytes, the sporophyte is not free living but attached to the photosynthetic gametophyte and derives nourishment from it.
Statement-II : In heterosporous pteridophytes the female gametophyte are retained on the parent sporophytes for variable periods.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

77) Read the following statements and select the correct option :
(a) Alcoholic fermentation occurs in muscles due to inadequate oxygen.
(b) Lactic acid fermentation involves removal of hydrogen from pyruvate.
(c) The main purpose of fermentation is to increase ATP production beyond glycolysis.
(d) Alcoholic fermentation and lactic acid fermentation both start with glycolysis.

(1) a, b, c are incorrect

(2) b & d are correct

(3) Only d is incorrect

(4) a, c, d are correct

78) **Statement-I :-** Proton concentration becomes higher in the intermembrane space than in the mitochondrial matrix during electron transport.
Statement-II :- The electron transport chain synthesizes ATP by substrate level phosphorylation.

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Both statement-I and statement-II are incorrect.

(4) Statement-I is incorrect but statement-II is correct.

79) During aerobic respiration, if oxygen supply to the cell suddenly stops, several metabolic processes will be affected. Considering the events in respiration, which process will stop first.

(1) Glycolysis

(2) Link reaction

(3) Kreb's cycle

(4) ETS

80) Removal of apical bud results in growth of lateral buds. Which plant hormone suppresses the growth of lateral buds ?

(1) Auxin

(2) Gibberellin

(3) Cytokinin

(4) Ethylene

81) How many among the following statements are **incorrect** ?
(a) Plant growth is defined as reversible increase in size or dry weight.
(b) Ethylene promotes fruit ripening and leaf senescence.
(c) Process by which cell lose their ability to divide and aquire specific functions is called differentiation.
(d) Abscisic acid promotes stomatal opening during water stress.
(e) Gibberellin induces bolting in rosette plants.

(1) Three

(2) Four

(3) Five

(4) Two

82) Which of the following is **not** a homopolymer?

(1) Chitin

(2) Inulin

(3) Cellulose

(4) Collagen

83) Match **List-I** with **List-II**

	List -I		**List-II**
A.	Codeine	I.	Polysaccharide
B.	Adenosine	II.	Alkaloid
C.	Inulin	III.	Nucleoside
D.	Trypsin	IV.	Protein

Choose the option with **all correct** matches.

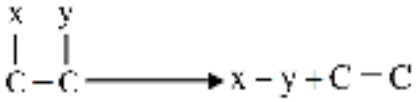
(1) A-II, B-III, C-IV, D-I

(2) A-III, B-II, C-IV, D-I

(3) A-III, B-II, C-I, D-IV

(4) A-II, B-III, C-I, D-IV

84) Name the class of enzyme that usually catalyze the following reaction:-



(1) Hydrolase

(2) Lyases

(3) Transferase

(4) Ligase

85) Carbon reactions of photosynthesis are dependent on which products of light reactions?

(1) ATP, NADPH, CO_2 and H_2O

(2) ATP and NADPH

(3) ATP, NADPH and O_2

(4) ATP, NADPH and CO_2

86) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In C_4 plants photorespiration does not occur.

Reason (R) : C_4 plants have a mechanism that increases the concentration of CO_2 at the RuBisCO enzyme's site i.e. mesophyll cell. This takes place when the C_4 acid is broken down in the mesophyll cells to release CO_2.

In the light of the above statements choose the most appropriate answer from the option given below.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

87) Find the **correct** statements:
A : In Calvin cycle the reduction step involves utilisation of 2 molecules of ATP for phosphorylation and two of NADPH for reduction per CO_2 molecule.
B : In all photosynthetic plants, the first stable product of CO_2 fixation in Calvin cycle is 3-phosphoglyceric acid.
C : Chemiosmosis require a membrane, a proton pump, a electron gradient and ATP synthase.
D : Biosynthetic phase 'or' dark reactions of photosynthesis are not light dependent.
E : It is probably to meet the difference in number of ATP and NADPH used in the dark reaction that the cyclic photophosphorylation takes place.
Choose the **correct** answer from the options given below:

(1) C and E only

(2) A, B, C and D only

(3) A, B and E only

(4) A and E only

88) Ciliated columnar epithelium is found in :-

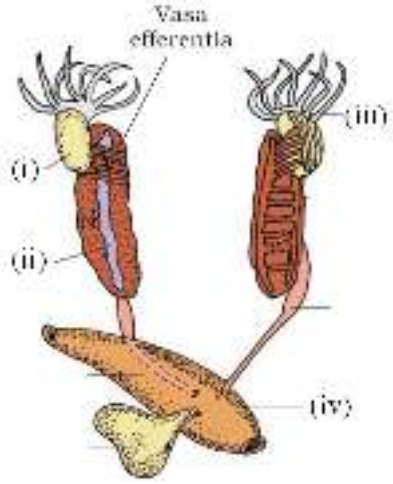
(1) Stomach & blood vessels

(2) Fallopian tube

(3) Proximal convulated tubule of nephron

(4) Airsac

89) Given below is the diagram of reproductive system of an animal labelled by (i), (ii), (iii) and (iv)



identify the animal with correct labelling of (i) to (iv)

(1) Male frog - (i) Testes (ii) Adrenal gland (iii) Fat bodies (iv) Cloaca

(2) Female frog - (i) Kidney (ii) Adrenal gland (iii) Ovary (iv) Cloaca

(3) Male frog - (i) Testes (ii) Duct of bidder (iii) Adrenal gland (iv) Cloaca

(4) Female frog - (i) Ovary (ii) Oviduct (iii) Adrenal gland (iv) Cloaca

90) (A) Cockroach are nocturnal omnivores that live in damp places throughout the world
(B) They become residents of human homes and thus are serious pests and vectors of several

diseases.
(C) Many species of cockroach are wild and are economic importance.
(D) They are pests, because they destroy food and contaminate it with their smelly excreta.
Select the incorrect statements regarding the cockroach

(1) A and B

(2) A, B and C

(3) B and D

(4) Only C

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	3	1	2	1	2	2	4	2	1	3	2	3	1	3	3	4	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	1	1	1	4	2	1	2	1	2	2	4	1	4	2	2	1	3	4
Q.	41	42	43	44	45															
A.	3	2	1	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	4	3	1	3	2	4	2	2	4	3	4	3	1	3	2	1	3	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	4	4	2	4	3	2	1	4	4	2	3	4	1	3	2	2	3	2	2
Q.	86	87	88	89	90															
A.	4	3	4	3	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	3	3	1	3	2	3	1	1	4	4	1	2	3	3	1	4	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	3	3	3	3	2	3	2	4	4	2	3	2	3	3	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	3	2	4	1	1	1	2	2	3	2	2	3	1	2	2	1	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	2	4	4	2	2	1	3	3	4	3	4	4	1	1	2	4	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	4	2	2	3	3	2	1	4										

SOLUTIONS

PHYSICS

1)

$$\lambda^0 = \frac{h}{mv_0}$$

$$v' = v^0 + \frac{q\varepsilon}{m}t \quad (q = e)$$

$$\lambda' = \frac{h}{m\left[v_0 + \frac{e\varepsilon_0}{m}t\right]}$$

$$\lambda' = \frac{h}{mv_0\left(1 + \frac{e\varepsilon_0}{mv_0}t\right)}$$

$$\lambda' = \frac{\lambda_0}{\left(1 + \frac{e\varepsilon_0}{mv_0}t\right)}$$

2) Change in angular momentum

$$\Delta L = L_2 - L_1 = \frac{n_2 h}{2\pi} - \frac{n_1 h}{2\pi}$$

$$= \frac{6.6 \times 10^{-34}}{2 \times 3.14} \times (5-4)$$

$= 1.05 \times 10^{-34}$ J.S

3) Effective gravity: $g_{eff} = g + a_0$

Tension in string between masses: $T_1 = \frac{2m_1 m_2}{m_1 + m_2} g_{eff}$

The pulley is pulled down by two tensions, so support tension: $T = 2T_1$

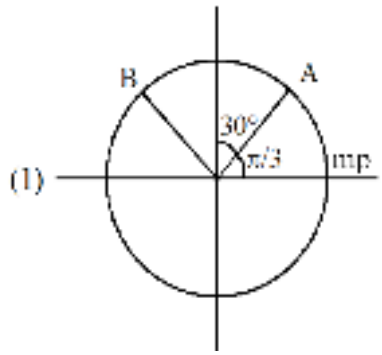
$$T = \frac{4m_1 m_2 (g + a_0)}{m_1 + m_2}$$

$$T = \frac{4\,[m]\,[3m]\,[g + g/5]}{m + 3m}$$

$$T = \frac{(4m)\,(3m)\,(6g/5)}{4m}$$

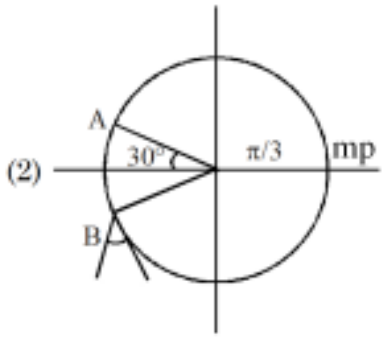
$$\Rightarrow T = \frac{18}{5}\ mg$$

4)



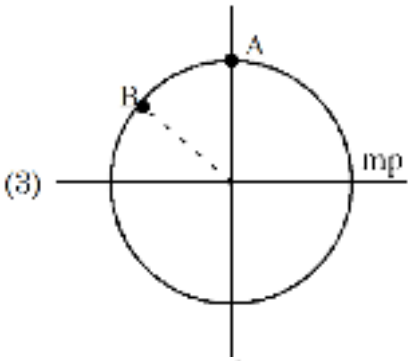
$\pi/3 = \omega t$

$t = \frac{\pi}{3\omega}$



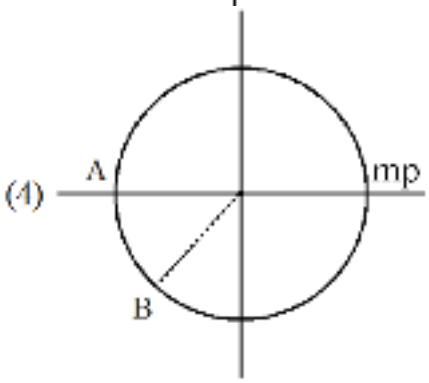
$\omega t = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

$t = \frac{5\pi}{6\omega}$



$\omega t = \frac{\pi}{2}$

$t = \frac{\pi}{2\omega}$



$\omega t = \pi$

$t = \frac{\pi}{\omega}$

5) Velocity at the bottom of the circle = $\sqrt{5g\ell}$

= $\sqrt{5 \times 10 \times 2}$ = 10 m/sec

on rough surface

u = 20 v = 10 a = – μg = – 5

$v^2 - u^2 = 2ax$

$10^2 - 20^2 = -2 \times 5 \times x$

x = 30 m

6)

Current through small element di= JdA

$$i = \int_{a}^{2a} \frac{J_0 2\pi r dr}{r^2}$$

$i = 2\pi J_0 \ln 2$

7) $I = 2.4\ kgm^2$, $K.E._{Rot.} = 750\ J$

$\alpha = 5\ rad/s^2$ $\omega_0 = 0\ rad\ s^{-1}$

$K.E. = \frac{1}{2} I\omega^2$; $\omega = \alpha t$

$$t = \sqrt{\frac{2K}{I\alpha^2}} = \sqrt{\frac{2 \times 750}{2.4 \times 25}} = 5\ sec$$

8) The intensity (I) of a wave depends on the square of its amplitude (A) and the square of its frequency (f), so $I \propto A^2 f^2$.

From the equations:

For wave 1: $A_1=5$, $f_1=75$

For wave 2: $A_2=10$, $f_2=150$

The ratio is

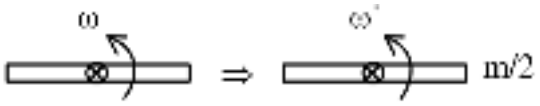
$$\frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2 \left(\frac{f_1}{f_2}\right)^2 = \left(\frac{5}{10}\right)^2 \left(\frac{75}{150}\right)^2$$

$$= \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}.$$

9) $\frac{\lambda}{2} = (58.5 - 21.0) \times 10^{-2} m$

$\lambda = 75 \times 10^{-2} m$,

$$f = \frac{v}{\lambda} = \frac{330}{75} \times 10^2 = 440 Hz$$

10) 

According to conservation of angular momentum,

$$\frac{1}{12} mL^2 . \omega = \left(\frac{1}{12} mL^2 + \frac{m}{2}\left(\frac{L}{2}\right)^2\right) \omega'$$

$$\omega' = \frac{2\omega}{5}\ rad/s$$

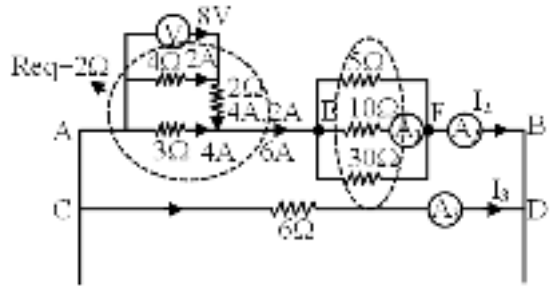
11) (a) New holes and conduction electrons are produced continuously throughout the material due to temperature.

(b) is correct as barriers is formed by immobile positive and negative ions left behind after mobile carriers combine. (c) is correct as biasing changes effective potential and temperature and doping affect intrinsic carrier concentration and built in voltage. (d) is incorrect as internal potential difference cannot be directly measured by an external voltmeter.

12) As the level of water in vessel decreases, the velocity of efflux will decrease. Hence it will

take more time for next 3 litre of water to come out.

13) Reading of voltmeter = 8V



Current through 4Ω and 2Ω resistor = 2A
Current through 3Ω resistor =(8 + 4)/3 = 4A
Current in upper Branch = I_2 = (4+2) = 6A ⇒ A_2
Req of 5Ω, 10Ω, 30Ω in parallel

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{5} + \frac{1}{10} + \frac{1}{30} \Rightarrow R_{eq} = 30$$

$V_{EF} = 6 \times 3 = 18V$
Current through 10Ω resistor

$$= \frac{18}{10} = 1.8A = I_1 \Rightarrow A_1$$

$$\frac{I_3}{I_2} = \frac{R_{AB}}{R_{CD}} \Rightarrow I_3 = \frac{I_2 R_{AB}}{R_{CD}} = \frac{6 \times 5}{6} = 5A \Rightarrow A_3$$

14) $$n_2 = n_1 \left[\frac{M_1}{M_2}\right] \left[\frac{L_1}{L_2}\right]^2 \left[\frac{T_1}{T_2}\right]^{-2}$$

$$= 1 \left[\frac{5kg}{1kg}\right] \left[\frac{20m}{m}\right]^2 \left[\frac{10s}{5}\right]^{-2} = 20J$$

15) $$x_n = \frac{3D\lambda}{a} = 5 \times 10^{-3}$$

$$\frac{3 \times \lambda \times 1}{3 \times 10^{-4}} = 5 \times 10^{-3}$$

$\lambda = 5 \times 10^{-7}$ m
λ = 5000 Å

16)

Potential energy stored in capacitor

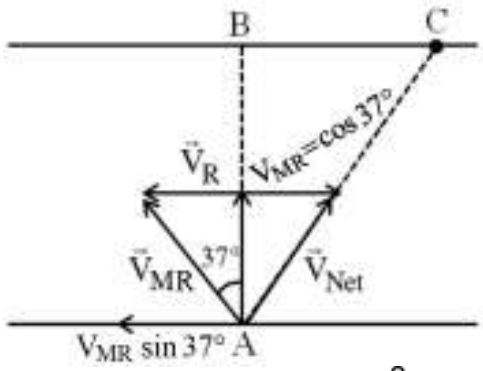
$$U = \frac{1}{2}CV^2$$

as battery is connected, it will keep the potential difference constant
So, U ∝ C
on insterting dielectric, C will increase
So U will also increase.
Also battery maintains the potential as constant.
So, if capacitance is varied, the value of charge will vary.

⇒ That means, Assertion is true but reason is a false statement.

17) $F = \frac{B^2 v \ell^2}{R} = \frac{4 \times 10^{-2} \times 1^2}{50} = 8 \times 10^{-4} N$

18)



As $V_X = V_{MR} \sin 37^o = 5 \times \frac{3}{5} = 3 km/h$

which is less river flow (V_R), so man will reach in the downstream point (C)

For crossing time

$t = \frac{AB}{V_{MR} \cos 37^o} = \frac{6}{\left(5 \times \frac{4}{5}\right)} = 1.5 hrs$

Now displacement of along the river flow

Shift = (V_R – V_{MR} sin 37°) t

Shift = (4 – 3) (1.5) = 1.5 km

19) $K_{min} = 0$

$K_{max} = \frac{hc}{\lambda} - \phi = \frac{12400}{4000} - 1.8 = 3.1 - 1.8 = 1.3$

K_{max} = 1.3 eV now work done by electric field

= 1.5 eV

so on reaching collector K'_{max} = 1.3+1.5 = 2.8eV

K'_{min} = 0 + 1.5 = 1.5 eV

20) $g' = \frac{g}{100}$

$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$

$\frac{g}{100} = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$

h = 9R

21)

Work done = W = q_0 (V_B–V_A) = q_0(V_C–V_A)

Here → $V_C - V_A = \int_{2a}^{3a} Edr$

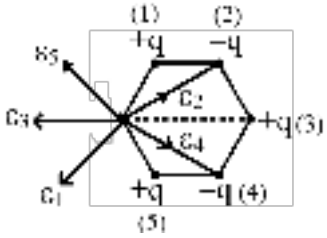
$= \int_{2a}^{3a} \frac{\lambda}{2\pi\varepsilon_0 r} dr = \frac{\lambda}{2\pi\varepsilon_0} \ell n \left(\frac{3}{2}\right)$

$\Rightarrow q_0 \frac{\lambda}{2\pi\varepsilon_0} \ell n \left(\frac{3}{2}\right)$



22) (P)

(Q) Direction would be along the line joining the center.



(R)

(S) At center ε-field would be zero.

23) $P = \frac{1}{2}\varepsilon_0 \left(\frac{V}{r}\right)^2$

Substitute values :

$P = \frac{1}{2}\left(8.85 \times 10^{-12}\right)\left(\frac{3000}{0.10}\right)^2$

$= \frac{1}{2}(8.85 \times 10^{-12})(9 \times 10^8) = 3.98 \times 10^{-3} N/m^2$

$\approx 4 \times 10^{-3}$ N/m^2

24)

If f > fr,
then $X_L > X_C$
Circuit nature is inductive.

25) **Rectifier:** Converts AC to DC, removing AC fluctuations.
Stabilizer: Ensures consistent voltage output regardless of input variations.
Transformer: Adjusts the voltage (step-up/step-down) based on the turns ratio.
Filter: Removes any ripple from the rectified DC signal by smoothing it.

26) $P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3} \times 8.99 \times 10^{-2} \times (3180)^2$

$= 3 \times 10^5 N/m^2 = 3$ atm

27)

Acc to kepler's law.

$$\frac{dA}{dt} = \frac{L}{2m}$$

$$dA = \frac{L}{2m} \times T$$

28) $\vec{r} = \left(t^2 - 4t + 6\right)\hat{i} + t^2\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = (2t - 4)\hat{i} + 2t\hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 2\hat{j}$$

$\vec{a} \cdot \vec{v} = 0$

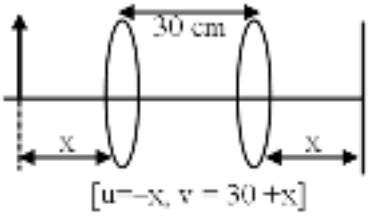
$4t - 8 + 4t = 0$

$t = 1$ sec

29) From displacement method.

$h_0 = \sqrt{h_{I_1} h_{I_2}} = \sqrt{16 \times 4} = 8\text{cm}$

For first position of lens :



[u=–x, v = 30 +x]

$$m_1 = \frac{h_{I_1}}{h_0} = \frac{-16}{8} = -2$$

$$m_1 = -2 = -\frac{(x + 30)}{x}$$

$\Rightarrow 2x = x + 30 \Rightarrow x = 30$ cm

$$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{60} - \frac{1}{-30}$$

$$\Rightarrow \frac{1+2}{60} = \frac{3}{60} \Rightarrow f = 20\text{cm}$$

30)

$$F_{AD} = \frac{\mu_0 I_1 I_2}{2\pi d_1} \times \ell \quad (d_1 = 5 \text{ cm})$$

$$F_{BC} = \frac{\mu_0 I_1 I_2}{2\pi d_2} \times \ell \quad (d_2 = 30 \text{ cm})$$

$F_{net} = F_{AD} - F_{BC}$

$F_{net} = 1.25 \times 10^{-4}$ N

31)

$A = A_0 e^{-Kt} \Rightarrow \ln A = -Kt + \ln A_0$

$y = mx + c \Rightarrow K = \frac{3\ell n2}{90} = \frac{\ell n2}{30}$

$\ell n(4) = -(K)t + \ell n(8)$

$Kt = 3\ell n2 - 2\ell n2 = \ell n2$

$= t = \frac{\ell n2}{K} = \frac{\ell n2}{\ell n2} \times 30 = 30$ s

32) Total energy of C^{12} nuclie

$= 12 \times 7.68$

$= 92.16$MeV

Similarly energy of C^{13} nuclie

$= 13 \times 7.47 = 97.11$ MeV

$\Delta E = (97.11 - 92.16) = 4.95$ MeV

33)

For a circular current carrying loop, magnetic filed at it's center & magnetic moment is given as.

$B = \frac{\mu_0 i}{2R}$ --------(1) {$A = \pi R^2$

$M = iA$ --------(2) {$R = \sqrt{\frac{A}{\pi}}$

Eq. (2)/Eq. (1)

$\frac{M}{B} = \frac{A\,2R}{\mu_0}$

$M = \frac{2BA}{\mu_0}\sqrt{\frac{A}{\pi}}$

34) $Y = \overline{\bar{A} + A.\bar{B}}$

$= \bar{\bar{A}}.\overline{(A.\bar{B})}$

$= A.(\bar{A} + \bar{\bar{B}})$

$= A.(\bar{A} + B)$

$= A.\bar{A} + A.B = A.B$

35) $i_d = \varepsilon_0 \frac{d\phi_E}{dt}$

$= \varepsilon_0 A \frac{dE}{dt}$

$= \frac{10^{-9}}{9 \times 4\pi} \times \pi \times (1.2)^2 \times \frac{200}{5}$

$= 1.6 \times 10^{-9}$ Amp

36) $P = KV$

PV^{-1} = constant

$C = \frac{f}{2}R + \frac{R}{1-x}$

$= \frac{5}{2}R + \frac{R}{2} = \frac{6R}{2} = 3R$

$Q = nC\,\Delta T$

$= 1 \times 3R\,(2T_0) = 6\,RT_0$

37) $\Delta n = n_1 - n_2 \Rightarrow 4$

$= \frac{v}{2l_1} - \frac{v}{2l_2} = \frac{v}{2}\left[\frac{1}{1.00} - \frac{1}{1.025}\right]$

$\Rightarrow v \simeq 328\,m/s$

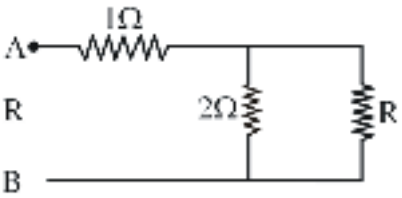
38) As Given , $F_{||} = 2100N$, $A = 100cm^2$, $\Delta x = 0.25$ cm , $L = 10$ cm

$$\eta = \frac{F_{||}/A}{\frac{\Delta x}{L}} = \frac{\frac{2100}{(10\times10)10^{-4}}}{\frac{0.25\times10^{-2}}{10\times10^{-2}}}$$

$$= \frac{21\times10^4}{25\times10^{-3}} = \frac{21}{25}\times10^7$$

$= 8.4 \times 10^6\ N/m^2$

39)



If $R_{AB} = R$

$\Rightarrow R = 1 + \frac{2R}{2+R}$

$\Rightarrow R(2+R) = 2 + R + 2R$

$\Rightarrow R^2 - R - 2 = 0$

$\Rightarrow R = 2\Omega$

40) $V_{com} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$

When both move opposition direction.

$V_{com1} = \frac{5\times1 - 2\times2}{1+2} = \frac{1}{3}$

when both move in same direction.

$V_{com2} = \frac{5\times1 + 2\times2}{1+2} = \frac{9}{3}$

$\frac{V_1}{V_2} = \frac{\frac{1}{3}}{\frac{9}{3}} = \frac{1}{9}$

41) 9 MSD = 10 VSD, 1 MSD = 0.1 cm, LC = 0.01cm

(a) When both jaws are closed : ZE is negative ;
MSR = 0 cm, VSR = – (10–4) × 0.01 cm ; ZE = – 0.06 cm
(b) During measurement : MSR = 3.5 cm ; VSR = 6 × 0.01 cm = 0.06 cm
Corrected reading of diameter = 3.5 cm + 0.06 cm + 0.06 cm = 3.62 cm

42)

General body : - A = $20m^2$; $e_r = 0.8$
t = 1min = 60sec ; $(Q)_{GB}$ = ?
Ideal black body : - $(I)_{IBB} = 200$ watt/m^2

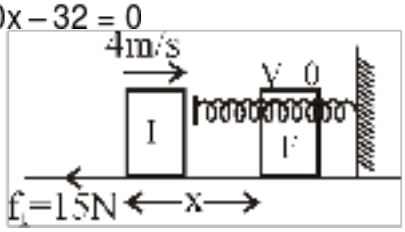
$e_r = \frac{(I)_{GB}}{(I)_{IBB}} \Rightarrow \frac{(I)_{GB}}{(I)_{IBB}} = 0.8$

$\Rightarrow (I)_{GB} = 0.8 \times (I)_{IBB}$
$\Rightarrow (I)_{GB} = 160$ watt/m^2
$\Rightarrow \frac{Q}{At} = 160$
$\Rightarrow Q = 160 \times A \times t$
$\Rightarrow Q = 160 \times 20 \times 60$
$\Rightarrow Q = 192000J \Rightarrow = 192kJ$

43) By WET
$W_{friction} + W_{spring} = \Delta KE$

$$-15(x) - \frac{1}{2}kx^2 = 0 - \frac{1}{2}mv^2$$

$$10000x^2 + 30x - 32 = 0$$



$\Rightarrow x = 5.5$ cm

44) $\frac{1}{f} = \left(\frac{\mu_1}{\mu_2} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

in air → $\frac{1}{15} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ...(1)

In medium → $\frac{1}{f_m} = \left(\frac{9}{8} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ...(2)

Now equation (1) divided by (2)

$$\frac{f_m}{15} = \frac{0.5}{\left(\frac{1}{8}\right)}$$

$f_m = 60$ cm

45) $i = \frac{E_{net}}{R_{net}} = \frac{1.5 - 1.3}{r_2 + r_1} = \frac{0.2}{r_1 + r_2}$

Reading of (V) = TPD of $E_1 \Rightarrow 1.45 = E + ir_1$

$\Rightarrow 1.45 = 1.3 + \left(\frac{0.2}{r_1 + r_2}\right) r_1$

$\Rightarrow \frac{0.2r_1}{r_1 + r_2} = 0.15 \Rightarrow 4r_1 = 3r_1 + 3r_2$

$\Rightarrow r_1 = 3r_2$

CHEMISTRY

46) NCERT, Class12th , Part-I , Article No : 3.4,
Pg.No : 79,80. Edition-2023-2024.

47) **Zero Order :** $r = K[A]^0$

First Order : Rate $\alpha [A]^1$, $t_{1/2} = \frac{0.693}{K}$

Second Order : Rate $\alpha [A]^2$, $t_{1/2} = \frac{1}{K[A_0]}$

Rate Doubling : If [A] is quadrupled and rate doubles $\Rightarrow$ rate $\alpha [A]^{0.5} \rightarrow$ (Half order).

48) $\pi_1 = \pi_2$

$(i_1 CRT) = i_2 (CRT)$

$i_1 = i_2$

for $Ca(NO_3)_2$ & $Na_2SO_4 \Rightarrow i = 3$

49) **(i) The vapor pressure of a liquid decreases with an increase in temperature.** This statement is *incorrect*. Vapor pressure increases with increasing temperature. Higher temperatures provide more kinetic energy to the liquid molecules, allowing them to escape into the gas phase more readily.

(ii) The liquid boils at the temperature at which its vapour pressure is equal to the atmospheric pressure.

This statement is *correct*. Boiling occurs when the vapor pressure of the liquid equals the surrounding pressure (usually atmospheric pressure).

(iii) Vapor pressure of the solvent decreases in the presence of a non-volatile solute.

This statement is *correct*. This is known as Raoult's Law. A non-volatile solute reduces the number of solvent molecules at the surface of the liquid, hindering their escape into the vapor phase and thus lowering the vapor pressure.

(iv) Vapor pressure of the pure solvent and solution is a function of temperature.

This statement is *correct*. As discussed in statement (i), temperature directly affects vapor pressure for both pure solvents and solutions.

50)

Given

$A^{2-} + B \rightarrow A + B^{2-}$

$E^0_{A^{2-}/A} > E^0_{B^{2-}/B}$

$B^{2-} + C \rightarrow$ No reaction

$E^0_{C^{2-}/C} > E^0_{B^{2-}/B}$
$C^{2-} + A \rightarrow C + A^{2-}$
$E^0_{C^{2-}/C} > E^0_{A^{2-}/A}$
Conclusion
$E^0_{C^{-2}/C} > E^0_{A^{-2}/A} > E^0_{B^{-2}/B}$

51) • **Incorrect** — PbO_2 is the **cathode**, not the anode.
• **Incorrect** — Electrolyte is **~38% H_2SO_4**, not 50%.
• **Correct** — During discharge, **PbO_2 → $PbSO_4$** at the **cathode**.
• **Incorrect** — On charging, **H_2SO_4 concentration increases**, not decreases.

52) Molality = $\frac{\text{moles of solute}}{\text{Mass of solvent (in kg)}}$
$= \frac{0.1}{0.5} = 0.2\,\text{m}$

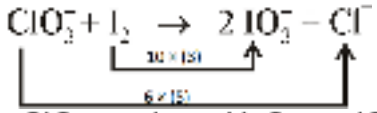
53) 0.1 mol NH_3 contain 0.1 mol N and 0.3 mol H atom
%H (by mass) = $\frac{3}{17} \times 100 = 17.65\%$
and no. of atoms = $0.1 \times 6.023 \times 10^{23} \times 4$

54) • de-Broglie wavelength: $\lambda = h / (mv)$
• Uncertainty principle: $\Delta x \cdot \Delta p \geq h / (4\pi)$
• Given: $\Delta x = \lambda / (4\pi)$
Solution:
$\Delta x \cdot \Delta p = (\lambda / 4\pi) \cdot \Delta p = h / 4\pi \Rightarrow \Delta p = h / \lambda$
But $\Delta P = m\Delta v = h / \lambda$; $\Delta v = \frac{h}{m\lambda} = \frac{h}{m\left(\frac{h}{mv}\right)}$
$\Delta v = v$

55) Original: $K_C = [Q]^2 [R]$
At new equilibrium,
$K_C = [2Q]^2 [R]' = 4[Q]^2 [R]'$
From both equation (I) and (II)
$[Q]^2 [R] = 4 [Q]^2 [R]'$
$\Rightarrow \quad [R]' = \frac{[R]}{4}$

56) For precipitation $[Zn^{2+}] [Cl^-]^2 > K_{sp}$.

57) $\overset{+5}{ClO_3^-} + \overset{0}{I_2} + xH_2O \rightarrow \overset{+5}{IO_3^-} + \overset{-1}{Cl^-} + yH^+$



$$5ClO_3^- + 3I_2 + 3H_2O \rightarrow 6IO_3^- + 5Cl^- + 6H^+$$

58) $\Delta H = \Delta U + \Delta n_g RT$

$= -90 - \frac{2 \times 2 \times 500}{1000}$

$= -92$ kcal/mole

$\Delta G = \Delta H - T\Delta S$

$= -92 \text{ kcal} - \frac{500(200)}{1000}$

$= -92 \text{ kcal} + 100 \text{ Kcal}$

$= +8$ kcal

equilibrium temp. $= \frac{\Delta H}{\Delta S} = \frac{92000}{200} = 460\,K$

59) By applying Hess law.

$\Delta H°_{solution} = \Delta H°_{lattice} + \Delta H°_{hydration}$

60)

$FeSO_4.(NH_4)_2SO_4.6H_2O + KMnO_4 \xrightarrow{H^+} Mn^{+2} + Fe^{+3}$

$Fe^{2+} \rightarrow Fe^{3+}$;

change in oxidation state = 1 → n-factor = 1;

Equivalent weight = 392 / 1 = 392

61) $\sqrt{\nu}$ v/s Z graph ⇒ straight line.

62) NCERT Page No. # 85

63)

Bond order $= \frac{1}{2}[N_b - N_a]$

H_2^+ **(1 electrons):** $\sigma 1s^1 \rightarrow$ BO = 0.5

He_2 **(4 electrons):** $\sigma 1s^2, \sigma^* 1s^2 \rightarrow$ BO = 0

N_2^+ **(13 electrons):** $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^1 \rightarrow$ BO = 2.5

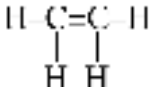
O_2^- **(17 electron):** $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^1 \rightarrow$ BO = 1.5

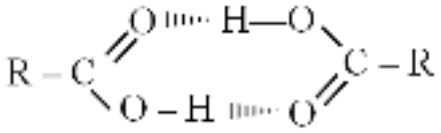
Hence, a–r, b–p, c–s, d–q.

Final answer :- (1)

64)

Multiple bond called as single super pair





H-bond

65)

66)

NCERT Pg. no. # 112
For greater covalent character of an ionic bond (i) small cation (ii) large anion.

67) **Concept:** EAN
Solution:

A. In sodium nitroprusside, $Na_2[Fe(CN)_5NO]$, the complex ion is $[Fe(CN)_5NO]^{2-}$.
B. The oxidation number of iron (Fe) is calculated as +2 (with NO considered as NO^+).
C. The coordination number of iron is 6 (five CN^- and one NO ligand).
D. The Effective Atomic Number (EAN) of iron is

The EAN formula is: EAN = Z - Oxidation State + Electrons from Ligands.
= 26 - 2 + 12 = 36.

68)

NCERT-XII, Pg. # 134

69) • (A) $[Pt(NH_3)_2Cl_2]$: Square planar → cis/trans geometrical isomerism → correct.
• (B) Optical isomers: Non-superimposable mirror images → called enantiomers → correct.
• (C) Chlorophyll: Contains magnesium in a porphyrin ring → correct.
• (D) Ethylenediamine (en): Bidentate ligand, not monodentate → incorrect.
Final Answer: Option (2)

70)

Concept :- Oxyacid's of Phosphorus
Solution :-
(1) Correct order of acids eq $H_3PO_4 < H_3PO_3 < H_3PO_2$
(2) H_3PO_4 → Oxidising agent
H_3PO_3 → Can act as both oxidising and reducing agent
H_3PO_2 → Reducing agent
(3) H_3PO_2 is monobasic oxide
(4) This is these statement as in all oxyacids phosphorus is tetrahedral in nature.
Answer :- 4
The geometry of phosphorus is tetrahedral in all the three

71)

Solution : C_{60} actually contains **12 pentagonal** and **20 hexagonal** rings, not the reverse. Hence, the given statement about ring count is incorrect.
Final Answer: Option (3)

72)

1. **Kernite:**
Kernite is a borate mineral with the chemical formula $Na_2B_4O_7 \cdot 4H_2O$.
This match is correct.
2. Borax:
Borax is another borate mineral with the chemical formula $Na_2B_4O_7 \cdot 10H_2O$. This match is incorrect. The correct formula for borax has 10 water molecules, not 8.
3. Cryolite:
Cryolite is a fluoride mineral with the chemical formula Na_3AlF_6.
4. Phosphinic Acid:
Phosphinic acid (also known as hypophosphorous acid) has the chemical formula H_3PO_2. This match is correct.
Answer:
Option : 2

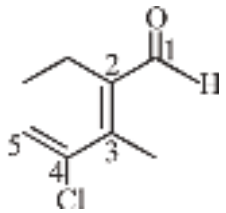
73) NCERT-XII, Pg. # 225
Fact for vanadium, Zn has fulfilled confi.

74)

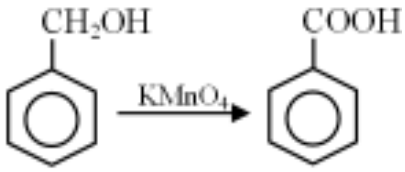
NCERT-XII, Pg. # 238

75)

(A) Lake Test → Al^{+3}
(B) Nessler's Reagent → NH_4^+
(C) Potassium sulphocyanide → Fe^{+3}
(D) Brown ring Test → NO_3^-

76)


4–Chloro–2–ethyl–3–methylpenta–2, 4–dienal

77) 

78)

1. Statement (1): Aniline is a Lewis base and reacts with the Lewis acid catalyst $AlCl_3$ (used in Friedel-Crafts) to form a deactivating salt. This prevents the reaction. This statement is Correct.
2. Statement (2): The $-NH_2$ group is highly activating and very susceptible to oxidation. Direct nitration with strong oxidizing acids (conc. HNO_3 + H_2SO_4) leads to the formation of tarry oxidation products. This statement is Correct.
3. Statement (3): In the strongly acidic nitration medium, aniline is protonated to form the anilinium ion ($C_6H_5NH_3^+$). This $-NH_3^+$ group is a deactivating, meta-directing group, leading to a substantial yield of m-nitroaniline. This statement is Correct.
4. Since statements (1), (2), and (3) are all correct, the correct option is (4).

79)

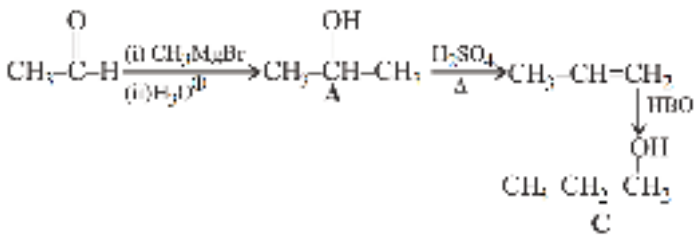
The reactions of benzenediazonium chloride can be matched as follows:
(A) The reaction of a diazonium salt with CuCl/HCl is the Sandmeyer reaction (iv), which yields chlorobenzene.
(B) The reaction with Cu/HCl is the Gattermann reaction (iii), which also forms chlorobenzene.
(C) The reaction is the Balz-Schiemann reaction (ii), which yields fluorobenzene.
(D) The reaction with phenol in an alkaline medium is a Coupling reaction (i), which forms an azo dye.
All these reactions are nucleophilic aromatic substitutions. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i).

80)

FACT

81)

All CH_3 groups are at maximum distance.

82) 

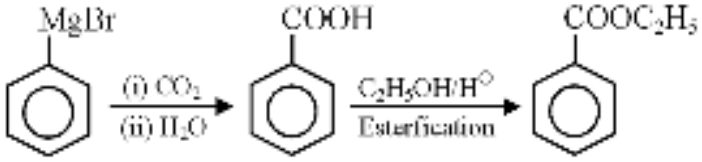
A & C are position isomer.

83)

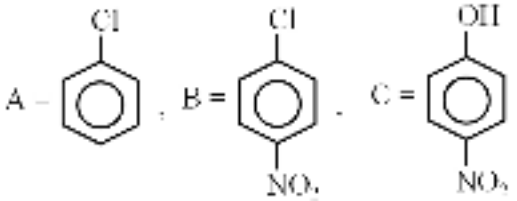
Benzylic substitution occurs by FRSR

84)

Answer Explanation:
Statement 1: Correct → Denaturation occurs by change in pH and temperature.
Statement 2: Incorrect → Denaturation does not affect the primary structure (sequence of amino acids).
Statement 3: Correct → Denaturation leads to loss of secondary and tertiary structure.
Statement 4: Correct → Sequence of amino acids remains unchanged.
Thus, the incorrect statement is loss of primary structure.

85) 

86) NCERT fact

87) 

88)

$CH_3 - Cl \xrightarrow[\text{Dry ether}]{Na} CH_3{-}CH_3$

$CH_3 - CH_2 - COONa \xrightarrow{\text{electrolysis}} CH_3{-}CH_3$

$CH_3{-}Cl \xrightarrow[2.CuCl \\ 3.CH_3-Cl]{1.Li} CH_3{-}CH_3$

$CH_3{-}Cl \xrightarrow[HCl]{Zn} CH_4$

89)

Presence of Chiral carbon.

90)

Bredt's Rule

BIOLOGY

91) NCERT-XII_Pg. # 136

92) NCERT-XII_Pg. # 131-133

93) NCERT Pg. # 139- 140

94) XII NCERT Pg. # 137

95) NCERT-XII_Pg. # 153-154

96) NCERT Pg. # 153, 155, 157, 158

97) NCERT Pg. # 151, 152

98)

NCERT Page No. # 114, 115

99)

NCERT Pg. # 123

100) NCERT-XII Pg. # 8

101) NCERT-XII Pg. # 11, 17, 20, 22

102) NCERT-XI_Pg. # 78,79

103) NCERT-XI_Pg. # 74-77

104) NCERT-XI_Pg. # 65

105) NCERT-XI_ Pg. # 62

106) NCERT XII_ Pg. # 19, 20

107)

NCERT-12th_Pg. # 72, 73

108) NCERT-12th Pg. # 69, 70, 71

109) NCERT-12th_ Pg # 53, 59, 62, 74

110) NCERT-12th_Pg. # 67, 68

111) NCERT 12th, Pg. # 241, 242, 244

112) NCERT-12th_ Pg. # 244, 245

113) NCERT-12th (Eng)_ Pg. # 245

114) NCERT 12th Pg. # 231

115) NCERT XII (Eng) pg. # 236

116) NCERT-XII_Pg. # 168

117) NCERT-XII_Pg. # 170-171

118) NCERT-XII_ Pg.# 173

119) NCERT (XII)_Pg. # 183-184

120) NCERT-XII_Pg. 180-181

121)

NCERT XII, Pg. # 83

122)

NCERT XII, Pg. # 90

123)

NCERT XII, Pg. # 96

124)

NCERT XII, Pg. # 101

125)

NCERT XII, Pg. # 99

126) NCERT Pg. # 108

127) NCERT-XII_Pg. # 42

128) NCERT-XII_Pg. # 27

129) NCERT-XII, Pg. # 37

130) NCERT-XII, Pg. # 44, 45

131) NCERT-XII_Pg. # 39

132) NCERT-XII, Pg. # 28

133) NCERT Pg # 97

134) NCERT Pg # 101

135) NCERT Pg # 99

136) NCERT 125 & 128

137) NCERT Pg # 123, 124 &127

138) NCERT Pg # 122

139) NCERT Pg. # 187

140) NCERT-Pg. # 186

141) NCERT_Pg. # 203

142) NCERT_Pg. # 197

143)

NCERT Pg. # 191

144) NCERT Pg. # 194

145)

NCERT Pg. # 196- 200

146) NCERT Pg. # 207

147) NCERT Pg. # 205, 207, 208, 210, 213

148) NCERT pg. # 208

149) NCERT Pg. # 223, 224, 225

150) NCERT Page No. # (E) 223, 224, 225 (H) 244, 245, 246

151) NCERT Page No. # (E) 233, 224, 225, (H) 244, 245, 246

152) NCERT XI, Pg. # 209

153)

NCERT XI, Pg. # 203

154)

NCERT XI, Pg. # 226

155)

Module

156)

NCERT XI Page No. # 42

157)

NCERT XI Page No. # 45

158)

NCERT XI Page No. # 41-44

159)

NCERT XI Page No. # 47

160)

NCERT XI Page No. # 48-51

161)

NCERT XI, Pg # 4

162)

NCERT XI Page No. # 14 & 15

163)

NCERT XI Page No. # 13, 17, 19, 21

164)

NCERT XI, Pg # 33

165)

NCERT XI, Pg # 27

166)

NCERT XI Page No. # 29, 32

167) NCERT Pg. # 157

168) NCERT-XI Pg. # 160

169) NCERT-XI Pg. 160

170) NCERT-XI Pg. # 175

171) NCERT XI, Pg. # 175,176,177

172) NCERT-XI_Pg. # 109, 110, 111

173) NCERT-XI_Pg. # 106, 108, 109, 110

174) NCERT-XI_Pg.# 117

175) NCERT-XI_Pg. # 136,142
हिंदी - Pg. # 137,172

176) NCERT-XI_Pg. # 147
हिन्दी_Pg. # 149

177)

NCERT XI, Pg. # 136,142-145

178) Old NCERT XI, Pg. # 102

179) New NCERT XI, Pg. # 83

180) Old NCERT XI, Pg. # 111 and 115